Supporting information: recovery, screening and trend inference

Author information to be completed by the authors

1 Repetition and inference units

The earlier single-year and assembly benchmark fixes twenty integer seeds, 128 conditional trajectories, twenty multiple imputations and 200 trend degradations. The new whole-archive benchmark instead uses twenty paired archive masks per station and 256 joint draws; its separate results and calibration support are reported below. In the leave-one-year-out source comparison, each outer region and all connected groups appearing in that region are excluded before forming fitting roles. For partitioned forward models, the original assignment excludes held test groups but can retain training-only groups in the same named region; the seed-zero audit and strict-region sensitivity below document the affected models. The remaining groups are assigned in shares 0.55/0.15/0.15/0.15 to estimation, interval calibration, failure learning and rule calibration; probability calibration is additionally separated inside failure learning. Bootstrap multiplicities stay inside roles. The source workflow requires at least four complete historical years. Case identity includes network, station, year, mask family, realization and reconstruction tier. The realization string alone is not unique.

Empirical repetition quantiles do not estimate uncertainty from new climates. The complete source data are the same in each repetition. Historical registered verdicts and all opened-data supplementary results retain their original status. No recalibration repairs a historical registered failure retrospectively.

There are 581 physical station identities but 602 network–station input instances, and 6,170 unique complete station-years versus 6,334 network–station–year tasks. Repeated physical records share their connected-group exclusion. Input identity therefore includes the network and its original time window. The observed-air control reconstructs those full-precision panels and original forcing, rather than using a clipped float32 archive proxy.

For AR(1) correlation ϕ and integer $m > 1$, the recovery model uses

$$v(m,\phi) = m^{-2}\{m + 2\sum_{j=1}^{m-1}(m-j)\phi^j\}, \qquad c(m,\phi) = (30m)^{-1}\sum_{i=1}^{m}\sum_{j=1}^{30}\phi^{|i-j|}.$$

The hidden interval in this covariance approximation is contiguous at one end of the window; correlations are clipped to $[-0.99, 0.999]$. The variance formula uses rounded hidden-day counts, and zero hidden share has zero component mean and variance before the numerical scale floor. These approximations do not constitute a distribution-free recovery bound. The stored nominal mixture scale combines untruncated component variances with truncated means; it is a diagnostic scale proxy, not the exact truncated-mixture variance. The Student sensitivity must be interpreted accordingly.

2 Numerical scores and censoring

For ensemble $x_1, \ldots, x_M$ and outcome y, CRPS is

$$M^{-1}\sum_{m}|x_m - y| - (2M^2)^{-1}\sum_{m,l}|x_m - x_l|.$$

Threshold-weighted scoring replaces both arguments by their maximum with the training H30 threshold, without selecting outcomes. Earlier historical analyses used midpoint quadrature with constant outer tails for seven stored quantiles. The revised main comparison applies a common observed-window support projection and a normal-spacing tail approximation to both methods; the tail is not identified by the seven quantiles. The whole-archive offsets are additionally re-estimated on the original independent interval-calibration role. Historical tables below retain their original estimates and must not be substituted for revised scores. The 90% interval score adds twenty times the distance outside the interval to its width.

With an annual observation known only to exceed L, the Gaussian trend contribution is $\log\Phi\{(\mu - L)/\sigma\}$. It is a survival likelihood for a lower bound. Such bounds can be informative because the observation window and annual maximum are related; this working model is evaluated as a comparator, not assumed correct. Entirely censored series and optimizer failures are unavailable, not exact trends.

3 Comparisons, availability and subgroup reporting

Point and distribution scorecards cross empirical calendars and controlled gaps with decision-time peak overlap and tier. Air-indicated hot conditions use training-role air quantiles. True heat and true peak overlap appear only as evaluation diagnostics. Pairwise station and group effect summaries require common available predictions. Risk tables retain total, admitted and evaluable-group counts. Unavailable scores are rejected; no admitted observations imply undefined selected risk.

Three neural splits contain the same held-case masks across MSE boosting, LSTM, SAITS and CSDI. Neural inputs are water anomalies, air anomalies and seasonal sine/cosine; neighbor series are omitted. Models use a 30-epoch CPU cap and early stopping on at most three evenly spaced validation years per group. Full case-weighted development runs were stopped for computational cost before CSDI test prediction; all final arms were retrained using the fixed group-balanced rule. CSDI computes H30 separately for twenty sampled trajectories. Matched annual-model and classical-baseline files use exactly these case identities and validation groups. Availability, including missing discharge, is determined before computing failure rates. Medians across three splits are descriptive and cannot satisfy the 18-of-20 abstract rule.

Individual caps and batch-average rules answer different questions. With within-group admitted and failed fractions A_g, B_g, the bounded-group certificate targets $E(B_g)/E(A_g)$ for uniform-group sampling. It bounds $B_g + r(1 - A_g)$ at r using independent bounded group units and a fixed candidate family. It provides no guarantee for every group and does not target the case-weighted ratio reported in ordinary scorecards. The eight-threshold family uses a Bonferroni allocation of the error budget. Even zero observed bounded loss requires approximately 99 groups for a 5% cap and 5% total error; a single threshold requires about 59. Rejecting every value provides no nonempty selected-risk evidence.

Overlapping-group adjustments, random-effects intervals, EM prior changes, domain weights, ATC and prediction-powered mean-risk estimates have their own unverified assumptions. The O/E analysis uses a group-unit sampling-variance floor rather than treating dependent simulated masks as independent events. The PPI

correction estimates a group-equal mean with an asymptotic interval; it is not a finite-sample semi-supervised selective-risk theorem. Synthetic label-shift controls change class priors while preserving class-conditional feature distributions; separate controls change gap, climate and conditional risk. The local-label curve reports nonempty certificates separately from realized risk.

4 Formal conditional and semi-supervised comparisons

The additional conditional-conformal implementation uses the authors' `conditionalconformal` 0.0.4 package and the binary LAC score $S(x, y) = |y - p(x)|$. Calibration samples one case uniformly per unique rule-calibration independence group, outside every held group. The basis contains overlapping macro-region, information-tier, peak-overlap, historical-heat, dam and zero-flow proxy indicators. The source rule-calibration role is distinct from the estimation, interval and failure-learning roles. Mask replicas are not additional calibration units.

Both the conservative and independently case-randomized versions are evaluated. The latter draws a test dual weight $u \sim \mathrm{Unif}(\tau - 1, \tau)$ separately for every case. The finite-basis boundary is computed by minimizing the calibration pinball loss tilted by $-u\Phi(x)^\top \beta$, then selecting the smallest target prediction among optimal fits. Mathematical unboundedness gives a full or empty prediction set according to the sign of the test weight. The boundary LP agrees with the authors' sensitivity algorithm in numerical tests, including random dual weights. Monotone cutoff caching retains casewise randomization and enlarges a set by at most the stated 10^{-8} numerical endpoint tolerance. Solver failures conservatively return the full binary set and are counted explicitly. An initial one-sided-residual development comparison is archived; the final binary-score results are separate.

For the nonrandomized version, every active binary group indicator requires at least $\tau/(1 - \tau)$ independent calibration representatives for a finite cutoff: nine at 90% and nineteen at 95%. This follows directly from the feasible range of the calibration dual sum for that group. It is a mathematical support limit, not a failed optimizer. The randomized version has a different support condition and is not suppressed by this nonrandomized count test. A previously unseen full region indicator permits only a full or empty randomized set, with no singleton release.

Group-conditional prediction-set coverage concerns $P(Y \in C(X) \mid G)$. Publishing only when $C = \{0\}$ therefore bounds false-admission frequency, not $P(Y = 1 \mid \text{published}, G)$. Both quantities are reported. The conditional theorem additionally assumes exchangeability of calibration and test representatives; source calibration and a held climatic region need not meet that condition.

The formal general-loss semi-supervised RCPS comparison applies the paper's Algorithms 1 and 4 to $L_g = B_g + r(1 - A_g)$. For distinct equally sized unlabeled blocks U_i,

$$W_i = |U_i|^{-1} \sum_{j \in U_i} \widetilde{L}_j + L_i - \widetilde{L}_i \in [-1, 2].$$

Finite-sample WSR bounds use predictable bets and stop a fixed testing sequence at its first failed candidate. Labeled, unlabeled-calibration and untouched test groups are disjoint. Too few distinct unlabeled groups is an explicit status. Labeled-only WSR and the earlier bounded-loss method remain separate comparators. Their target is a same-population group-equal selected-risk ratio under independent identically distributed group sampling, rather than every river's conditional risk.

5 Frozen-output deletion sensitivity and observed record checks

Group-deletion sensitivity freezes predictions and original admissions. Risk crossings reduce to additive integer margins $db_g - na_g$ for rational cap n/d. Retention reversals solve a binary integer program requiring that both remaining rules satisfy the same point-risk cap and the reference remains nonempty. The published comparator is selected once on opened full-data outcomes. A zero deletion count means no initial qualified advantage; impossible reversals are undefined. Source checks cover all twenty seeds and retain witness group lists. Historical N1 reproduces its original 7% point-risk endpoint, not the later proposed 6%/8% E1 criteria. Reapplying the batch rule at a frozen-admission witness is reported separately and has no claimed minimum-deletion optimality.

The observation-only record check fixes 1981–2025 and requires at least fifteen complete observed years at a station. It recomputes every complete reference H30 and every incomplete-year bound from the original daily inputs. A witness uses thirty consecutive finite observations, with dates and all thirty values retained. An observed-window mean above the maximum of all complete-year H30 values

disproves the complete-only record attribution for the reported series. It does not require the candidate to be the unique hottest year, nor assert zero sensor error. A 5×10^{-6} degree numerical tolerance removes floating-point ties. Regional sensitivity separately fixes rosters and first-to-last-complete-year spans; common-year and expanded-year rankings distinguish added support from changes among already eligible years.

6 Supplementary evidence and remaining dependencies

The full machine-readable tables, seed-level results, deep training records, forcing provenance, 7DADM qualification, natural-reference exclusion reasons and New Zealand catalog request are repository artifacts. The natural search yields no qualified difficult annual peak-window outage among 180 candidate pair-years. A zero-failure binomial calculation concerns independent gaps, not dependent daily measurements. Twenty-three station-years have median air/donor residual correlation 0.505. Shared primary-sensor error can induce that correlation; it neither identifies separate sensor errors nor establishes three independent error channels. The reference-precision table assumes Gaussian AR1 errors and bounds linear window means; it does not measure logger accuracy.

The prior opened low-flow illustration is retained below. It concerns April–March positive-flow records and annual seven-day minimum discharge, reconstructed through log-flow regression and an annual distribution model. Its source files are `scripts/461_tr2_flow_demo.py` and `results/wrr_target_recovery_20261022/s1_flow`. A separate matched extension reproduces all 134,925 historical cases and adds best-donor MOVE.1, the same single-donor Ridge, and twenty Gaussian AR1 log-flow bridges. These working bridges use other-year in-sample residuals without independent interval calibration. In 33,609 controlled dangerous donor cases, log-scale MAE is 0.090 for single-donor Ridge and 0.106 for MOVE.1; the multidonor bridge gives 0.104 and historical D2 gives 0.075. The variance-preserving methods can change bias direction without improving MAE. Exponentiating a log-flow prediction differs from a conditional mean in original discharge units. This is descriptive prior evidence, not independent confirmation or a claim that every reconstructed extreme shares one empirical bias sign.

Cold-to-hot and calendar splits refit upstream quantities after removing all hot or future years, including from Air2stream, anomaly reference and donor regression training. The annual non-air stress is a deliberately incompatible diagnostic. The separate full daily perturbation adds 0.2°C per decade to all complete targets and donors and repeats upstream fitting under three seeds. These test different assumptions. Daily-year shifts are applied consistently to complete records and empirical mask calendars.

The Daymet comparison across all 319 USGS source sites changes both spatial product and daily averaging convention. Discharge uses the same original air features with log-flow lags; contradictory daily flow records remain unavailable. All 67,316 original POWER control identities and available points are reproduced exactly. Each alternative has its own shared-case reference. The earlier 35-site standardized-Ridge comparison remains a separate sensitivity. Lags now refer to actual calendar days, including missing leap-year dates, and every unavailable case remains explicit. The prior positional-lag results were reconstructed from the archived script and unchanged cached data; their original output bytes were not retained before overwrite. The European acquisition has 16 usable stations of 16 requested, and 4474 paired cases including 458 dangerous air-ridge cases. Nearby GHCN-Daily TAVG matching covers all 319 benchmarkable source USGS sites: 147 have a metadata-eligible station within 25 km, and 122 supply common valid cases. The exact original source control reproduces 67,316 case identities, availability and every available annual point. NOAA daily source S ends at 2400 UTC, while other source conventions vary; this is an observed-input sensitivity, not a perfect riverbank-air or clock-matched oracle. Missing inputs remain in all denominators. Daily metrics use identical dates in both arms. The 7DADM analysis uses approved maxima at 34 sites, without regulatory certification. Historical GAGES-II dam metadata are joined by exact station identifier; zero-flow fractions are proxies and lake-outlet classification is limited to one explicit agency designation. The distance screen fails to flag Big Sandy Creek; observed errors do not determine exclusion.

The selection exercise concerns an assembled panel rather than all USGS sites. Complete-year versus retained-year station eligibility is 98 versus 255 overall and 36 versus 163 in the USGS panel. Same-station slope contrasts, region-group bootstrap intervals, station BH adjustments, synchronized year permutations and

circular shifts are reported separately. Estimated incomplete outcomes cannot prove the truth of a changed scientific conclusion. Hot-year hit rates, false alarms and SEDI are preserved even when no valid event denominator exists. The original specification table varies tolerances and decision caps on fixed predictions. A separate twenty-seed, 540-outer-fit H7 pipeline recomputes seven-day peak support, masks, histories, functional draws, MOVE augmentation and annual/risk calibration. Native H7 fields are adapted to the legacy fitting engine after removing the genuine secondary H30 fields; exported native predictions restore H7 names. The daily-input audit reconstructs sixteen provider/tier combinations directly, and default H30 parity passes 240 cases and 201 fields. Deep models and all tolerance/model combinations are not refitted.

The Greater Wellington intake now contains actual native observations from all nine selected sites. Code 600 remains the specified accepted quality code; the returned legacy quality codes supply no eligible complete-reference hot years under that criterion. Consequently the external qualification denominator is nine and the recovery-accuracy denominator is zero. The study is unregistered. The source model and classical interval corrections were not adapted to obtain favorable target errors. Measurement-code compatibility must be resolved before treating these observations as reference truth.

The unflagged-exposure experiment uses the same source cases, masks and true annual targets in both conditions. Seven visible warm days are replaced by concurrent air readings, using air and visibility alone to locate the episode. About 97.6% of cases support such an episode, with an average 2.3°C daily air-minus-water shift on replaced days. This is an imposed stress severity, not an estimate of field exposure prevalence. Clean source weights and interval offsets remain fixed; the three seeds vary masks and trajectories. Table 31 separates missing-component and total trend coverage, and Table 32 reports unchanged-score release. Censored trends are omitted here because a contaminated visible window is no longer a valid physical lower bound. Unflagged exposure in an otherwise defined water target must not be interpreted as measurable water temperature in a dry bed.

Forcing-specific recovery probabilities are also recomputed from residual histories excluding both the target year and each residual-validation year. All 269,264

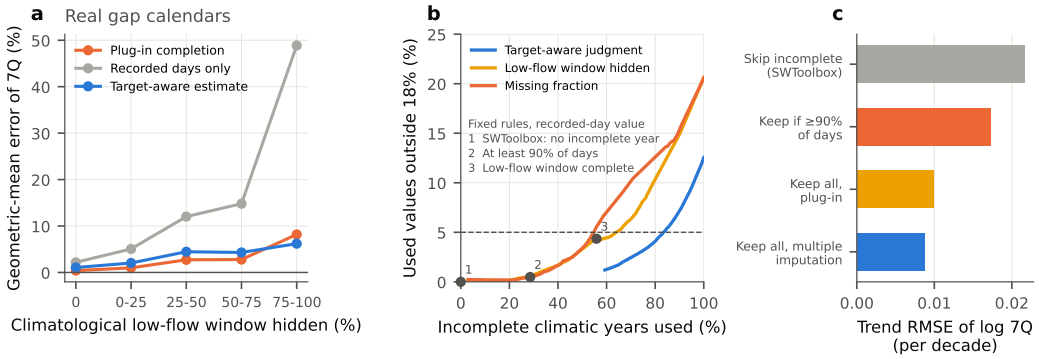


Figure 1. Previously opened low-flow illustration, preserved as supplementary context. Its log-flow estimators and incomplete-year practices have their own assumptions and are not the new twenty-seed thermal benchmark.

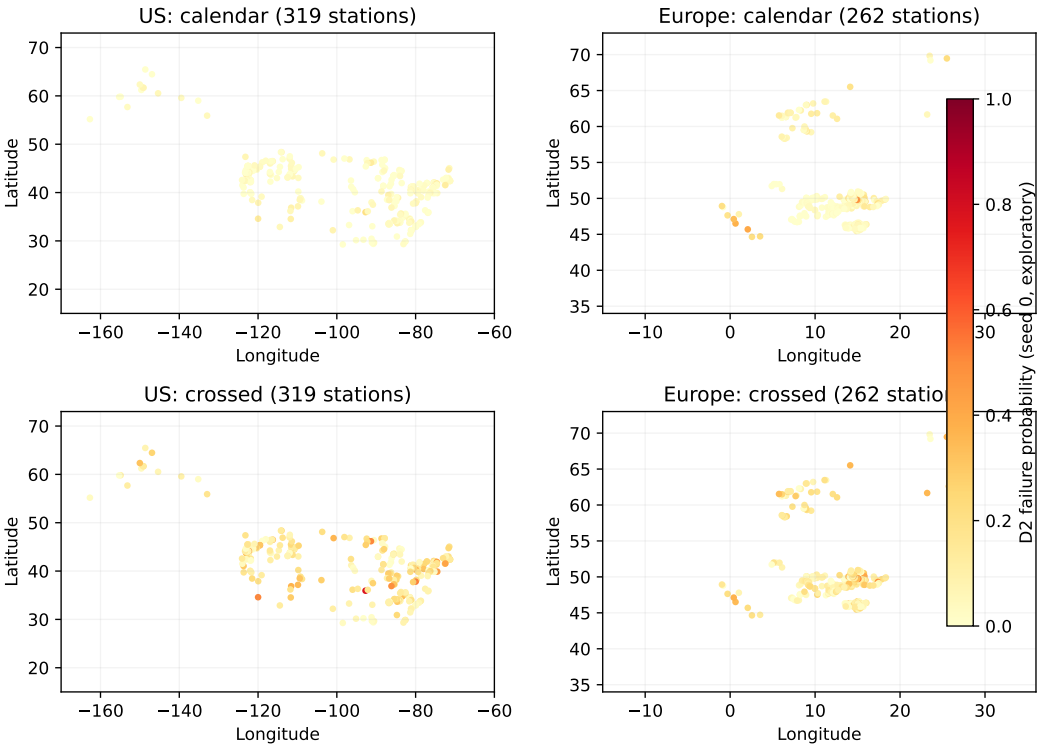


Figure 2. Opened-source station failures and available metadata. Coordinates are available for all 581 stations. The map supports descriptive failure analysis, not causal river-type attribution.

USGS arm/case points reproduce their existing controls, and 1,083 independent daily Ridge checks agree within numerical tolerance. In controlled dangerous air-tier cases, raw Gaussian recovery probabilities average 0.405 against observed failure 0.468 for POWER, and 0.365 against 0.457 for POWER plus flow. These have different available supports and failure targets; they do not establish paired calibration improvement or safe transfer. Complete scores and calibration bins are retained in CSV. Annual distribution and release-classifier weights remain unchanged.

The two stations specified in the review plan, USGS 13342500 and 06054500, each contribute twenty complete evaluation years; their median shares of random-missingness A1 squared error are 13.0 and 6.1 percent. Historical dam and zero-flow metadata do not identify thermal causation. Named European 2003, 2018 and 2022 examples use common finite cases across all four estimators and preserve event contingency counts, hit/false-alarm rates and undefined SEDI. The 2003 dangerous calendar sample has only 6–22 cases per repetition; median MAE is 0.913 degrees C for D2 and 0.595 for MOVE.1. US illustration years are selected using positive regional air anomalies with at least three complete-water stations, preserving unavailable regions. These opened examples do not establish universal estimator rankings.

The C1 stratification retains the original 336-USGS-site, 6,479-station-year missingness sample. Approved discharge yields 21 sites with recorded zeros, 266 without a recorded zero and 49 with unavailable flow. A station-conditional logistic model adjusts standardized summer air temperature for linear decade trend; 400 independence-group bootstrap repetitions give OR 1.208 [0.882, 1.564] for the zero-record group and 1.058 [0.988, 1.134] for the available-flow group without recorded zeros. Requiring thirty zero days gives 16 sites and OR 1.282 [0.876, 1.889]. Flow-adjusted specifications have smaller available samples and do not establish mediation. Zero flow is a limited historical proxy, and group exchangeability remains assumed.

The national extension fixes one canonical mean-temperature series per catalog site before numerical analysis, without combining sensors. All 1,273 candidates are attempted; 340 retain unresolved eligibility. National CSVs give eligibility bounds, one-degree station-grid occupancy, HUC2 counts, trends, basin-block stability, BH

and synchronized field tests. Hot-year methods share a station threshold from the 90th percentile of at least five complete approved 1981–2010 H30 values; insufficient references remain unavailable. Same-roster/span and common-year comparisons separate added support from changed values. Observed windows provide hot-frequency lower bounds and record witnesses without imputation. Conditional imputation ranges are model sensitivities, not validated confidence coverage.

7 Whole-archive supplements

Full paired MSE cells, group and region sensitivities, unavailable cases and source calibration support are in the review-response results. Forward task calibration ends before 2010. Normal-spacing tail extrapolation is compared to constant tails under the same observed-window constraint; its outer distribution is assumed, not identified by the seven stored quantiles. Auxiliary rank coupling is not a posterior for the annual estimator.

8 Additional USGS count-only candidate screen

An additional metadata-only roster of 10 stations was fixed before requesting 2010–2024 approved daily-mean counts. All retained unresolved river-network identities; no temperatures or reconstruction errors were saved. No station had at least four full years in the initial screen. After that result, the same entire roster was checked for 2025; this extension is post hoc and unregistered. No station is confirmed to pass the four-year count requirement, and 1 remains undetermined because the 2025 statistic was not returned. Three other missing 2025 statistics cannot change eligibility because their original counts are at most two. For the undetermined candidate, the current metadata end precedes 2025, but metadata may lag observations and alternative series were not screened. Count completeness does not establish nonestimated daily observations, complete subdaily coverage, sensor accuracy, river-network independence or climate independence. No candidate was scored.

Table 1. Historical registered verdicts. The redesign cannot revise them.

endpoint	verdict
P1	not met
P2	not met
P3	not met
P4	met historically
N1–N3	met after recomputation

Table 2. Exploratory direction stability; positive differences favor distribution regression. MAE and trend entries are fractional reductions; CRPS entries are score differences in degrees C. These are repeated analyses of the same archive.

metric	comparison	positive	repeats	median
MAE reduction	Daily plug-in	20	20	0.212
MAE reduction	Functional debiasing	20	20	0.053
MAE reduction	Lookup debiasing	20	20	0.202
MAE reduction	Annual MOVE.1	20	20	0.263
MAE reduction	Warm-season MOVE.1	20	20	0.243
MAE reduction	Stochastic regression	6	20	-0.003
MAE reduction	Calibrated stochastic	11	20	0.002
MAE reduction	Calibrated state	20	20	0.072
CRPS improvement	Calibrated state	20	20	0.017
Tail CRPS improvement	Calibrated state	20	20	0.004
CRPS improvement	State simulation	20	20	0.018
Tail CRPS improvement	State simulation	20	20	0.004

Table 3. Exploratory direction stability; positive differences favor distribution regression. MAE and trend entries are fractional reductions; CRPS entries are score differences in degrees C. These are repeated analyses of the same archive.

metric	comparison	positive	repeats	median
CRPS improvement	Stochastic regression	12	20	0.001
Tail CRPS improvement	Stochastic regression	0	20	-0.001
CRPS improvement	Calibrated stochastic	7	20	-0.001
Tail CRPS improvement	Calibrated stochastic	0	20	-0.001
Trend RMSE reduction	mcar	20	20	0.695
Trend RMSE reduction	non air trend	0	20	-0.602
Trend RMSE reduction	strong heat	20	20	0.696

Table 4. Observed-window record attribution on reported daily series, with no imputation. The eligible population contains 96 stations and at least fifteen complete observed years per station in 1981–2025. Positive excess means the complete-only record cannot be the full-record maximum.

margin c	n station years	n stations	n groups
0.000	35	20	15
0.500	12	6	6
1.000	3	2	2

Table 5. scorecard; source tables contain all strata and empirical repetition intervals.

family	estimator	seeds	MAE	CRPS	coverage
calendar	H30 MOVE.1	20	0.992	–	–
calendar	Annual MOVE.1	20	0.550	–	–
calendar	Warm-season MOVE.1	20	0.509	–	–
calendar	Air2stream-3	20	0.568	–	–
calendar	Calibrated state	20	0.465	0.335	0.875
calendar	State simulation	20	0.465	0.337	0.803
calendar	Distribution regression	20	0.451	0.331	0.863
calendar	Functional debiasing	20	0.476	–	–
calendar	Lookup debiasing	20	0.553	–	–
calendar	Observed days	20	4.108	–	–
calendar	Daily plug-in	20	0.564	–	–
calendar	Stochastic regression	20	0.463	0.343	0.723
calendar	Calibrated stochastic	20	0.463	0.338	0.853
calendar	triage_hybrid	20	0.451	–	–
crossed	H30 MOVE.1	20	1.009	–	–
crossed	Annual MOVE.1	20	0.453	–	–
crossed	Warm-season MOVE.1	20	0.441	–	–
crossed	Air2stream-3	20	0.534	–	–
crossed	Calibrated state	20	0.363	0.264	0.898
crossed	State simulation	20	0.363	0.264	0.824

Table 6. scorecard; source tables contain all strata and empirical repetition intervals.

family	estimator	seeds	MAE	CRPS	coverage
crossed	Distribution regression	20	0.337	0.248	0.888
crossed	Functional debiasing	20	0.355	–	–
crossed	Lookup debiasing	20	0.422	–	–
crossed	Observed days	20	1.427	–	–
crossed	Daily plug-in	20	0.428	–	–
crossed	Stochastic regression	20	0.336	0.248	0.753
crossed	Calibrated stochastic	20	0.337	0.247	0.892
crossed	triage_hybrid	20	0.337	–	–

Table 7. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
calendar	Distribution regression	all	20	1.000	0.030	0.071
calendar	Distribution regression	atc	20	0.965	0.014	0.006
calendar	Distribution regression	Batch average	20	0.994	0.027	0.033
calendar	Distribution regression	Batch + individual cap	20	0.966	0.014	0.011
calendar	Distribution regression	gap_49	20	0.806	0.005	0.000
calendar	Distribution regression	Group certificate	20	0.000	–	–
calendar	Distribution regression	individual_0.05	20	0.938	0.005	0.000
calendar	Distribution regression	Individual score ≤0.10	20	0.948	0.008	0.000
calendar	Distribution regression	individual_0.2	20	0.959	0.011	0.006
calendar	Distribution regression	individual_0.3	20	0.967	0.014	0.011
calendar	Distribution regression	Probability + group adjustment	20	0.921	0.003	0.000
calendar	Distribution regression	mondrian	20	0.966	0.014	0.011
calendar	Distribution regression	none	20	0.000	–	–
calendar	Distribution regression	one_feature_25	20	0.889	0.003	0.000
calendar	Distribution regression	Peak overlap <50%	20	0.916	0.005	0.000
calendar	Distribution regression	published_Beaufort_full	20	0.645	0.001	0.000
calendar	Distribution regression	published_Beaufort_full_as_written	20	0.641	0.001	0.000
calendar	Distribution regression	published_Dunham_full	20	0.654	0.002	0.000
calendar	Distribution regression	published_Dunham_full_as_written	20	0.655	0.002	0.000
calendar	Distribution regression	published_ETCCDI_15_3	20	0.350	0.000	0.000

Table 8. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
calendar	Distribution regression	published_GSIM_350	20	0.590	0.000	0.000
calendar	Distribution regression	published_NorWest90	20	0.847	0.004	0.000
calendar	Distribution regression	published_NorWest90_as_written	20	0.848	0.004	0.000
calendar	Distribution regression	published_Oregon80	20	0.873	0.003	0.000
calendar	Distribution regression	published_Oregon80_as_written	20	0.875	0.003	0.000
calendar	Distribution regression	published_USGS_311_24	20	0.517	0.000	0.000
calendar	Distribution regression	published_USGS_strict7	20	0.167	0.000	0.000
calendar	Distribution regression	published_WMO_11_5	20	0.432	0.000	0.000
calendar	Distribution regression	published_complete_year	20	0.000	–	–
calendar	Distribution regression	published_gap14	20	0.648	0.001	0.000
calendar	Distribution regression	published_gap49	20	0.806	0.005	0.000
calendar	Distribution regression	published_warm_gap14	20	0.813	0.001	0.000
calendar	Distribution regression	published_warm_gap30	20	0.885	0.003	0.000
calendar	Distribution regression	published_warm_gap7	20	0.745	0.000	0.000
calendar	Distribution regression	random_effects	20	0.000	0.003	0.000
calendar	Distribution regression	90% warm-season coverage	20	0.789	0.000	0.000
calendar	Observed days	one_feature_25	20	0.889	0.009	0.000
calendar	Observed days	Peak overlap <50%	20	0.916	0.018	0.000
calendar	Observed days	published_Beaufort_full	20	0.645	0.001	0.000
calendar	Observed days	published_Beaufort_full_as_written	20	0.641	0.002	0.000

Table 9. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
calendar	Observed days	published_Dunham_full	20	0.654	0.004	0.000
calendar	Observed days	published_Dunham_full_as_written	20	0.655	0.004	0.000
calendar	Observed days	published_ETCCDI_15_3	20	0.350	0.000	0.000
calendar	Observed days	published_GSIM_350	20	0.590	0.003	0.000
calendar	Observed days	published_NorWest90	20	0.847	0.011	0.000
calendar	Observed days	published_NorWest90_as_written	20	0.848	0.012	0.000
calendar	Observed days	published_Oregon80	20	0.873	0.010	0.000
calendar	Observed days	published_Oregon80_as_written	20	0.875	0.010	0.000
calendar	Observed days	published_USGS_311_24	20	0.517	0.001	0.000
calendar	Observed days	published_USGS_strict7	20	0.167	0.000	0.000
calendar	Observed days	published_WMO_11_5	20	0.432	0.000	0.000
calendar	Observed days	published_complete_year	20	0.000	–	–
calendar	Observed days	published_gap14	20	0.648	0.009	0.000
calendar	Observed days	published_gap49	20	0.806	0.025	0.000
calendar	Observed days	published_warm_gap14	20	0.813	0.008	0.000
calendar	Observed days	published_warm_gap30	20	0.885	0.018	0.000
calendar	Observed days	published_warm_gap7	20	0.745	0.002	0.000
calendar	Daily plug-in	one_feature_25	20	0.889	0.005	0.000
calendar	Daily plug-in	Peak overlap <50%	20	0.916	0.008	0.000
calendar	Daily plug-in	published_Beaufort_full	20	0.645	0.001	0.000

Table 10. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
calendar	Daily plug-in	published_Beaufort_full_as_written	20	0.641	0.002	0.000
calendar	Daily plug-in	published_Dunham_full	20	0.654	0.003	0.000
calendar	Daily plug-in	published_Dunham_full_as_written	20	0.655	0.003	0.000
calendar	Daily plug-in	published_ETCCDI_15_3	20	0.350	0.000	0.000
calendar	Daily plug-in	published_GSIM_350	20	0.590	0.001	0.000
calendar	Daily plug-in	published_NorWest90	20	0.847	0.006	0.000
calendar	Daily plug-in	published_NorWest90_as_written	20	0.848	0.007	0.000
calendar	Daily plug-in	published_Oregon80	20	0.873	0.005	0.000
calendar	Daily plug-in	published_Oregon80_as_written	20	0.875	0.005	0.000
calendar	Daily plug-in	published_USGS_311_24	20	0.517	0.001	0.000
calendar	Daily plug-in	published_USGS_strict7	20	0.167	0.000	0.000
calendar	Daily plug-in	published_WMO_11_5	20	0.432	0.000	0.000
calendar	Daily plug-in	published_complete_year	20	0.000	–	–
calendar	Daily plug-in	published_gap14	20	0.648	0.003	0.000
calendar	Daily plug-in	published_gap49	20	0.806	0.010	0.000
calendar	Daily plug-in	published_warm_gap14	20	0.813	0.003	0.000
calendar	Daily plug-in	published_warm_gap30	20	0.885	0.007	0.000
calendar	Daily plug-in	published_warm_gap7	20	0.745	0.001	0.000
crossed	Distribution regression	all	20	1.000	0.126	0.662
crossed	Distribution regression	atc	20	0.861	0.078	0.384

Table 11. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
crossed	Distribution regression	Batch average	20	0.858	0.078	0.368
crossed	Distribution regression	Batch + individual cap	20	0.847	0.074	0.346
crossed	Distribution regression	gap_49	20	0.333	0.082	0.400
crossed	Distribution regression	Group certificate	20	0.000	–	–
crossed	Distribution regression	individual_0.05	20	0.708	0.038	0.068
crossed	Distribution regression	Individual score ≤0.10	20	0.768	0.052	0.149
crossed	Distribution regression	individual_0.2	20	0.832	0.070	0.303
crossed	Distribution regression	individual_0.3	20	0.871	0.081	0.405
crossed	Distribution regression	Probability + group adjustment	20	0.618	0.021	0.027
crossed	Distribution regression	mondrian	20	0.841	0.072	0.311
crossed	Distribution regression	none	20	0.000	–	–
crossed	Distribution regression	one_feature_25	20	0.381	0.015	0.014
crossed	Distribution regression	Peak overlap <50%	20	0.492	0.035	0.043
crossed	Distribution regression	published_Beaufort_full	20	0.281	0.003	0.005
crossed	Distribution regression	published_Beaufort_full_as_written	20	0.291	0.013	0.030
crossed	Distribution regression	published_Dunham_full	20	0.281	0.007	0.006
crossed	Distribution regression	published_Dunham_full_as_written	20	0.292	0.011	0.011
crossed	Distribution regression	published_ETCCDI_15_3	20	0.000	–	–
crossed	Distribution regression	published_GSIM_350	20	0.000	–	–
crossed	Distribution regression	published_NorWest90	20	0.436	0.042	0.086

Table 12. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
crossed	Distribution regression	published_NorWest90_as_written	20	0.442	0.043	0.114
crossed	Distribution regression	published_Oregon80	20	0.285	0.006	0.005
crossed	Distribution regression	published_Oregon80_as_written	20	0.313	0.015	0.022
crossed	Distribution regression	published_USGS_311_24	20	0.000	–	–
crossed	Distribution regression	published_USGS_strict7	20	0.000	–	–
crossed	Distribution regression	published_WMO_11_5	20	0.000	–	–
crossed	Distribution regression	published_complete_year	20	0.000	–	–
crossed	Distribution regression	published_gap14	20	0.000	–	–
crossed	Distribution regression	published_gap49	20	0.333	0.082	0.400
crossed	Distribution regression	published_warm_gap14	20	0.181	0.001	0.006
crossed	Distribution regression	published_warm_gap30	20	0.478	0.058	0.238
crossed	Distribution regression	published_warm_gap7	20	0.159	0.001	0.000
crossed	Distribution regression	random_effects	20	0.000	0.000	0.000
crossed	Distribution regression	90% warm-season coverage	20	0.184	0.001	0.006
crossed	Observed days	one_feature_25	20	0.381	0.038	0.024
crossed	Observed days	Peak overlap <50%	20	0.492	0.097	0.397
crossed	Observed days	published_Beaufort_full	20	0.281	0.005	0.000
crossed	Observed days	published_Beaufort_full_as_written	20	0.291	0.022	0.044
crossed	Observed days	published_Dunham_full	20	0.281	0.013	0.000
crossed	Observed days	published_Dunham_full_as_written	20	0.292	0.024	0.039

Table 13. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
crossed	Observed days	published_ETCCDI_15_3	20	0.000	–	–
crossed	Observed days	published_GSIM_350	20	0.000	–	–
crossed	Observed days	published_NorWest90	20	0.436	0.114	0.557
crossed	Observed days	published_NorWest90_as_written	20	0.442	0.121	0.595
crossed	Observed days	published_Oregon80	20	0.285	0.013	0.000
crossed	Observed days	published_Oregon80_as_written	20	0.313	0.034	0.070
crossed	Observed days	published_USGS_311_24	20	0.000	–	–
crossed	Observed days	published_USGS_strict7	20	0.000	–	–
crossed	Observed days	published_WMO_11_5	20	0.000	–	–
crossed	Observed days	published_complete_year	20	0.000	–	–
crossed	Observed days	published_gap14	20	0.000	–	–
crossed	Observed days	published_gap49	20	0.333	0.305	0.984
crossed	Observed days	published_warm_gap14	20	0.181	0.000	0.003
crossed	Observed days	published_warm_gap30	20	0.478	0.213	0.962
crossed	Observed days	published_warm_gap7	20	0.159	0.000	0.000
crossed	Daily plug-in	one_feature_25	20	0.381	0.021	0.022
crossed	Daily plug-in	Peak overlap <50%	20	0.492	0.049	0.141
crossed	Daily plug-in	published_Beaufort_full	20	0.281	0.004	0.006
crossed	Daily plug-in	published_Beaufort_full_as_written	20	0.291	0.017	0.049
crossed	Daily plug-in	published_Dunham_full	20	0.281	0.008	0.008

Table 14. rules; source tables contain all strata and empirical repetition intervals.

family	delivery	rule	seeds	retention	risk	groups >10%
crossed	Daily plug-in	published_Dunham_full_as_written	20	0.292	0.015	0.033
crossed	Daily plug-in	published_ETCCDI_15_3	20	0.000	–	–
crossed	Daily plug-in	published_GSIM_350	20	0.000	–	–
crossed	Daily plug-in	published_NorWest90	20	0.436	0.057	0.200
crossed	Daily plug-in	published_NorWest90_as_written	20	0.442	0.059	0.232
crossed	Daily plug-in	published_Oregon80	20	0.285	0.008	0.011
crossed	Daily plug-in	published_Oregon80_as_written	20	0.313	0.020	0.043
crossed	Daily plug-in	published_USGS_311_24	20	0.000	–	–
crossed	Daily plug-in	published_USGS_strict7	20	0.000	–	–
crossed	Daily plug-in	published_WMO_11_5	20	0.000	–	–
crossed	Daily plug-in	published_complete_year	20	0.000	–	–
crossed	Daily plug-in	published_gap14	20	0.000	–	–
crossed	Daily plug-in	published_gap49	20	0.333	0.127	0.641
crossed	Daily plug-in	published_warm_gap14	20	0.181	0.001	0.006
crossed	Daily plug-in	published_warm_gap30	20	0.478	0.089	0.454
crossed	Daily plug-in	published_warm_gap7	20	0.159	0.001	0.000

Table 15. recovery; source tables contain all strata and empirical repetition intervals.

family	scope	model	seeds	auc median
calendar	all	lookup	20	0.921
calendar	all	one_feature	20	0.933
calendar	all	oracle_D2	20	1.000
calendar	all	oracle_plugin	20	1.000
calendar	all	recovery_model	20	0.972
calendar	all	recovery_t5	20	0.969
calendar	all	risk_classifier	20	0.978
calendar	danger	lookup	20	0.643
calendar	danger	one_feature	20	0.615
calendar	danger	oracle_D2	20	1.000
calendar	danger	oracle_plugin	20	1.000
calendar	danger	recovery_model	20	0.780
calendar	danger	recovery_t5	20	0.767
calendar	danger	risk_classifier	20	0.751
calendar	within_cell	recovery_model	20	0.873
crossed	all	lookup	20	0.785
crossed	all	one_feature	20	0.756
crossed	all	oracle_D2	20	1.000
crossed	all	oracle_plugin	20	1.000
crossed	all	recovery_model	20	0.867

Table 16. recovery; source tables contain all strata and empirical repetition intervals.

family	scope	model	seeds	auc median
crossed	all	recovery_t5	20	0.856
crossed	all	risk_classifier	20	0.866
crossed	danger	lookup	20	0.635
crossed	danger	one_feature	20	0.557
crossed	danger	oracle_D2	20	1.000
crossed	danger	oracle_plugin	20	1.000
crossed	danger	recovery_model	20	0.784
crossed	danger	recovery_t5	20	0.771
crossed	danger	risk_classifier	20	0.763
crossed	within_cell	recovery_model	20	0.806

Table 17. Matched deep-comparator crossed cases with dangerous gaps and air-ridge information tier; three seeds only. Neighbor data are absent from neural inputs.

model	seeds	MAE	bias median	availability median
boosting	3	0.556	0.008	1.000
Global lagged-air boosting	3	0.490	0.105	1.000
Local air boosting	3	0.447	0.097	1.000
Local flow boosting	3	0.398	0.129	0.600
Global flow boosting	3	0.579	0.257	0.600
CSDI	3	0.531	-0.010	1.000
distribution_regression_calibrated	3	0.333	0.155	1.000
distribution_regression_raw	3	0.333	0.155	1.000
Functional debiasing	3	0.338	0.133	1.000
LSTM	3	0.479	0.135	1.000
SAITS	3	0.460	-0.003	1.000
stochastic_regression_calibrated_local	3	0.261	-0.065	1.000
stochastic_regression_local	3	0.261	-0.065	1.000

Table 18. Matched deep-comparator calendar cases with dangerous gaps and air-ridge information tier; three seeds only. Neighbor data are absent from neural inputs.

model	seeds	MAE	bias median	availability median
boosting	3	0.793	-0.397	1.000
Global lagged-air boosting	3	0.699	-0.045	1.000
Local air boosting	3	0.643	-0.320	1.000
Local flow boosting	3	0.345	-0.065	0.429
Global flow boosting	3	0.620	-0.281	0.429
CSDI	3	0.689	-0.304	1.000
distribution_regression_calibrated	3	0.470	0.069	1.000
distribution_regression_raw	3	0.470	0.069	1.000
Functional debiasing	3	0.524	0.134	1.000
LSTM	3	0.697	-0.083	1.000
SAITS	3	0.825	-0.518	1.000
stochastic_regression_calibrated_local	3	0.894	-0.526	1.000
stochastic_regression_local	3	0.894	-0.526	1.000

Table 19. Nonempty bounded-group certification by local label budget. Undefined selected risks are not zero-risk successes.

family	labels (groups)	runs	nonempty certificate fraction	mean retention	risk among nonempty
calendar	2	40000	0.000	0.000	–
calendar	4	28000	0.000	0.000	–
calendar	8	18000	0.000	0.000	–
calendar	10	14000	0.000	0.000	–
calendar	16	6000	0.000	0.000	–
crossed	2	40000	0.000	0.000	–
crossed	4	28000	0.000	0.000	–
crossed	8	18000	0.000	0.000	–
crossed	10	14000	0.000	0.000	–
crossed	16	6000	0.000	0.000	–

Table 20. Poststratified outcomes per 1,000 historically supported incomplete station-years. Empirical repetition medians; unsupported strata stay in the denominator.

rule	used per 1000 median	failures per 1000 median	unsupported population per 1000 median
all	997.162	28.588	2.838
individual_0.05	933.980	3.997	2.838
Individual score ≤0.10	942.808	5.769	2.838
individual_0.2	953.531	9.070	2.838
individual_0.3	961.355	11.766	2.838
one_feature_25	893.668	2.913	2.838
Peak overlap <50%	924.923	4.721	2.838
warm90	799.642	0.583	2.838

Table 21. Exact frozen-admission deletion sensitivity. The best published comparator is fixed using opened full-data outcomes at a 5 percent point-risk cap. A zero is an initial failure; unavailable reversal minima remain undefined.

family	rule	repeats	finite minima	initial failures	median min. groups
calendar	Batch average	20	20	0	179.500
calendar	Batch + individual cap	20	20	0	168.500
calendar	Individual score ≤0.10	20	20	0	150.500
calendar	Probability + group adjustment	20	20	0	87.000
crossed	Batch average	20	20	20	0.000
crossed	Batch + individual cap	20	20	20	0.000
crossed	Individual score ≤0.10	20	19	15	0.000
crossed	Probability + group adjustment	20	20	0	114.000

Table 22. Historical opened N1 uses a 7 percent point-risk criterion. Fixed-admission minima and batch-rule reapplication at the witness answer different questions; historical verdicts do not change.

population	n groups	minimum groups	remaining risk	reapplied witness risk
deploy	40	10	0.071	0.056
mech	40	10	0.071	0.062

Table 23. Conditional conformal prediction-set comparators over twenty seeds. Conditional coverage targets false-admission frequency per eligible representative, not selected risk. Region means are descriptive under held-region shift; empty releases have undefined selected risk.

family	method	nominal coverage	seeds	retention	coverage	selected risk	LP fallback (%)
calendar	Coarse groups, conservative	0.900	20	0.000	1.000	–	0.000
calendar	Coarse groups, conservative	0.950	20	0.000	1.000	–	0.000
calendar	Coarse groups, randomized	0.900	20	0.156	0.900	0.005	1.754
calendar	Coarse groups, randomized	0.950	20	0.078	0.950	0.005	0.962
calendar	Full region, randomized	0.900	20	0.000	0.901	–	0.000
calendar	Full region, randomized	0.950	20	0.000	0.950	–	0.000
calendar	Full region, conservative	0.900	20	0.000	1.000	–	0.000
calendar	Full region, conservative	0.950	20	0.000	1.000	–	0.000
crossed	Coarse groups, conservative	0.900	20	0.000	1.000	–	0.000
crossed	Coarse groups, conservative	0.950	20	0.000	1.000	–	0.000
crossed	Coarse groups, randomized	0.900	20	0.111	0.900	0.083	0.824
crossed	Coarse groups, randomized	0.950	20	0.055	0.950	0.086	0.510
crossed	Full region, randomized	0.900	20	0.000	0.900	–	0.000
crossed	Full region, randomized	0.950	20	0.000	0.950	–	0.000
crossed	Full region, conservative	0.900	20	0.000	1.000	–	0.000
crossed	Full region, conservative	0.950	20	0.000	1.000	–	0.000

Table 24. Identical classifier and cap: isolate Platt probability calibration from its additional empirical overlapping-group factors. These are exploratory source repetition medians, not the formal conditional-conformal algorithm.

family	method	retention	risk	fraction groups above 10	worst group risk
calendar	Probability + group adjustment	0.921	0.003	0.000	0.045
calendar	Platt probability calibration	0.925	0.003	0.000	0.047
calendar	Raw classifier	0.948	0.008	0.000	0.080
crossed	Probability + group adjustment	0.618	0.021	0.027	0.159
crossed	Platt probability calibration	0.638	0.024	0.032	0.170
crossed	Raw classifier	0.768	0.052	0.149	0.231

Table 25. General-loss semi-supervised and labeled-only RCPS with WSR bounds. The unit is an independent group. Distinct unlabeled calibration groups and untouched test groups are required.

method	labels (groups)	runs	nonempty	unlabeled shortage
Labeled-only WSR	2	40000	0.000	0.000
Labeled-only WSR	4	28000	0.000	0.000
Labeled-only WSR	8	18000	0.000	0.000
Labeled-only WSR	10	14000	0.000	0.000
Labeled-only WSR	16	6000	0.000	0.000
Semi-supervised WSR	2	40000	0.000	0.400
Semi-supervised WSR	4	28000	0.000	0.643
Semi-supervised WSR	8	18000	0.000	1.000
Semi-supervised WSR	10	14000	0.000	1.000
Semi-supervised WSR	16	6000	0.000	1.000

Table 26. General-loss semi-supervised and labeled-only RCPS with WSR bounds. The unit is an independent group. Distinct unlabeled calibration groups and untouched test groups are required.

method	labels (groups)	runs	nonempty	unlabeled shortage
Labeled-only WSR	2	40000	0.000	0.000
Labeled-only WSR	4	28000	0.000	0.000
Labeled-only WSR	8	18000	0.000	0.000
Labeled-only WSR	10	14000	0.000	0.000
Labeled-only WSR	16	6000	0.000	0.000
Semi-supervised WSR	2	40000	0.000	0.400
Semi-supervised WSR	4	28000	0.000	0.643
Semi-supervised WSR	8	18000	0.000	1.000
Semi-supervised WSR	10	14000	0.000	1.000
Semi-supervised WSR	16	6000	0.000	1.000

Table 27. Same-roster and same-span regional sensitivity, 1981–2025. Annual means require 80 percent fixed-roster support. Common-year and expanded-year rankings differ. Regions with fewer than three independent groups have no bootstrap interval.

region	n stations	n groups	common years	top complete	top retained	trend difference
provider:chmi	11	4.000	18.000	2015.000	2015.000	0.000
provider:foen	11	2.000	25.000	2018.000	2018.000	0.009
provider:gkd_bayern	37	8.000	10.000	2022.000	2018.000	-0.009
provider:usgs	36	19.000	1.000	2020.000	2018.000	-0.046
region:usgs_huc02	5	2.000	10.000	2002.000	2002.000	0.111
region:usgs_huc03	5	4.000	11.000	2011.000	2010.000	-0.015
region:usgs_huc04	5	2.000	12.000	2020.000	2012.000	-0.178
region:usgs_huc17	15	7.000	8.000	2015.000	2015.000	0.027

Table 28. Observed-window record certificates among 96 stations with at least fifteen complete observed years in 1981–2025. Each incomplete-year lower bound exceeds every complete-year H30 at the same station; it need not be the unique full-record maximum.

margin c	n station years	n stations	n groups
0.000	35	20	15
0.500	12	6	6
1.000	3	2	2

Table 29. Source-only joint E1/E2 planning across all twenty algorithm and group-composition repetitions. Each budget has 40,000 empirical group simulations. Seed quantiles do not represent independent-climate uncertainty; E1 uses a planning delta approximation.

independent groups	n source seeds	simulation repetitions per seed	total simulations	joint E1 E2 probability	source seed lower	source seed upper
20	20	2000	40000	0.389	0.186	0.661
30	20	2000	40000	0.337	0.118	0.640
40	20	2000	40000	0.306	0.071	0.639
60	20	2000	40000	0.253	0.040	0.631
80	20	2000	40000	0.216	0.020	0.630
100	20	2000	40000	0.192	0.010	0.623
150	20	2000	40000	0.153	0.003	0.614
200	20	2000	40000	0.134	0.000	0.609

Table 30. Exploratory redesigned P4 analogues. Original numeric criteria require RMSE reduction at least 20 percent, bias no worse by over 0.01 degrees per decade, and missing-component coverage 90–97 percent. Total interval coverage cannot substitute for missing-component coverage. Historical P4 is unchanged.

scenario	repeats	passing	missing coverage median	total coverage median	rmse reduction median
mcar	20	0	0.792	1.000	0.695
non_air_trend	20	0	0.037	1.000	-0.602
strong_heat	20	0	0.796	1.000	0.696

Table 31. Unflagged visible-air exposure: medians of three mask/trajectory seeds with clean source-fit seed zero held fixed. Missing-component and total coverage refer to 95 percent trend intervals. Air replaces seven consecutive visible warm-day water readings; the underlying target remains defined. Random and strongly heat-associated missingness have matched marginals. Censored trends are omitted because contaminated windows are no longer valid lower bounds. This is synthetic measurement contamination, not dry-bed water truth or twenty-fit stability.

condition	scenario	pipeline	rmse	bias	missing coverage	total coverage
Clean	Random	90% warm-season rule	0.039	0.001	–	1.000
Clean	Random	Distribution-regression MI	0.011	-0.001	0.796	1.000
Clean	Random	Calibrated stochastic MI	0.012	-0.001	0.721	1.000
Clean	Heat	90% warm-season rule	0.045	-0.001	–	1.000
Clean	Heat	Distribution-regression MI	0.014	-0.000	0.799	1.000
Clean	Heat	Calibrated stochastic MI	0.014	-0.001	0.719	1.000
Air	Random	90% warm-season rule	0.085	-0.004	–	0.997
Air	Random	Distribution-regression MI	0.075	-0.004	0.169	0.998
Air	Random	Calibrated stochastic MI	0.075	-0.005	0.135	0.998
Air	Heat	90% warm-season rule	0.099	0.014	–	0.993
Air	Heat	Distribution-regression MI	0.091	0.017	0.161	0.995
Air	Heat	Calibrated stochastic MI	0.090	0.016	0.130	0.995

Table 32. The same clean source failure classifier and individual cap 0.1 under unflagged air exposure. Three mask/trajectory seeds share one source fit. Group risks require at least twenty admissions; no contaminated labels recalibrate the score.

condition	family	retention	risk	groups >10%	worst group risk
Clean	calendar	0.946	0.007	0.000	0.065
Clean	crossed	0.766	0.049	0.168	0.224
Air	calendar	0.950	0.369	0.818	1.000
Air	crossed	0.792	0.379	0.919	0.952

Table 33. Matched forcing sensitivities at all 319 benchmarkable USGS source sites. MAE is in degrees C. Exact original panels, forcing and daily Ridge code reproduce all 67,316 POWER control identities and available points. Each alternative has its own shared-case reference; different rows must not be compared as if they used one sample. Missing inputs stay in eligible denominators. Daymet uses a minimum/maximum midpoint; daily time conventions also vary. Nearby TAVG is not a perfect co-located or clock-matched oracle.

Input	Stratum	Eligible	Paired	Stations	POWER MAE	Alternative MAE
Daymet	All	67316	66133	319	0.317	0.325
Daymet	Danger	27530	27418	319	0.640	0.657
Daymet	Air, danger	9347	9271	308	0.661	0.682
POWER + flow	All	67316	51060	261	0.334	0.302
POWER + flow	Danger	27530	20972	259	0.683	0.612
POWER + flow	Air, danger	9347	7048	244	0.708	0.623
NOAA TAVG	All	67316	22028	122	0.318	0.312
NOAA TAVG	Danger	27530	8984	118	0.655	0.642
NOAA TAVG	Air, danger	9347	3010	112	0.678	0.671

Table 34. Native H7 source refits, 20 seeds and 540 outer-region fits. Seven-day peak supports, masks, history summaries, conditional functionals, MOVE augmentation, annual models and disjoint calibration are recomputed. MAE and proper scores are in degrees C; coverage refers to 90 percent intervals. These are dangerous decision-time cases, not the historical H30 endpoint; deep models are not refitted here.

family	estimator	seeds	MAE	CRPS	twCRPS	coverage
calendar	Distribution regression	20	0.494	0.365	0.065	0.874
calendar	Functional debiasing	20	0.522	–	–	–
calendar	Daily plug-in	20	0.638	–	–	–
calendar	Calibrated stochastic	20	0.516	0.380	0.068	0.858
crossed	Distribution regression	20	0.348	0.257	0.046	0.892
crossed	Functional debiasing	20	0.375	–	–	–
crossed	Daily plug-in	20	0.423	–	–	–
crossed	Calibrated stochastic	20	0.360	0.265	0.048	0.893

Table 35. The same 35 observation-only complete-year witnesses under each screening rule. The accepted-record comparison uses observed windows, not unknown complete H30 values. The 49-day and July–August rows are contextual proxies, not published H30 rules.

Rule	Witnesses	Admitted	Excluded	Above accepted
Complete	35	0	35	35
Annual 95	35	31	4	3
Station warm 90	35	32	3	2
Fixed warm 90	35	32	3	2
July–August	35	30	5	3
Gap $\leq$49 d	35	34	1	1

Table 36. Seven-station stress comparisons on common available station/repetition rows. Heat-associated masks and numbers of missing years differ between arms; unavailable cases remain in the eligible denominator. Coverage and width are supplied in the result table.

Scenario	Eligible	Available	Base RMSE	Stress RMSE
Donors / forward	30	30	0.079	0.116
Donors / offline	35	35	0.037	0.047
Heat / forward	30	30	0.079	0.088
Heat / offline	35	30	0.035	0.058

Table 37. Release followed by finite-record trend inference for controlled gaps. The values-and-bounds interval is a Gaussian working-likelihood interval, including regression variability; it is not calibrated missing-data uncertainty.

Mode	Released data	Eligible	Available	RMSE	Working width
forward	Values + bounds	820	820	0.202	3.262
forward	Values	820	820	0.210	–
offline	Values + bounds	840	840	0.040	1.986
offline	Values	840	840	0.043	–

Table 38. Task calibration and independent-draw intervals on exactly the same supported cases. Coverage is a fraction; missing source support is reported separately.

Setting	Method	n	Original coverage	Task coverage	Width ratio
calendar / forward	D2	40	0.925	0.975	0.947
calendar / forward	Stochastic	38	1.000	1.000	0.773
calendar / offline	D2	95	0.937	0.958	1.167
calendar / offline	Stochastic	88	0.977	0.989	1.225
crossed / forward	D2	39	1.000	1.000	1.112
crossed / forward	Stochastic	38	1.000	1.000	1.015
crossed / offline	D2	301	0.977	0.980	1.236
crossed / offline	Stochastic	307	0.948	0.951	1.108

Table 39. Task calibration support. Empirical calibration from small source pools does not establish a finite-sample guarantee.

Family	Time setting	Estimator	Availability	Minimum source groups
calendar	forward	D2	0.049	3
calendar	forward	stochastic_calibrated	0.046	3
calendar	offline	D2	0.113	3
calendar	offline	stochastic_calibrated	0.105	3
crossed	forward	D2	0.048	3
crossed	forward	stochastic_calibrated	0.046	3
crossed	offline	D2	0.358	3
crossed	offline	stochastic_calibrated	0.365	3

Table 40. Additional observation-only witnesses from approved USGS daily means. Every candidate remains in the qualification table. Sparse complete-reference support precludes an incidence or transfer inference.

Station	Year	Complete n	Observed L	Complete max	Excess
08377900	2016	3	14.990	14.853	0.137
08377900	2018	3	15.537	14.853	0.683
08377900	2020	3	15.737	14.853	0.883
08405200	2013	2	28.693	28.327	0.367
08405200	2015	2	28.703	28.327	0.377
08405200	2016	2	28.537	28.327	0.210
08405200	2019	2	28.833	28.327	0.507
08405200	2020	2	28.970	28.327	0.643
07364133	2015	1	29.427	28.767	0.660

Table 41. Replay of annual interval stages on common finite-bound cases. Raw is the original model output; stochastic raw quantiles use a finite set of imputations that retain observed days. First applies the first explicit bound projection; Pre-final applies the fixed projection-aware offsets and sorts; Final applies the bound again. New is the count first becoming zero-width at the final projection. Zero widths use 10^{-8} °C tolerance. These are repeated mask cases, and no alternative calibration was fitted.

Family	Setting	Estimator	n	Raw	First	Pre-final	Final	New
calendar	forward	D2	3674	0	2	1096	1476	380
calendar	forward	Stochastic	3674	3090	3090	2977	2977	0
calendar	offline	D2	6919	0	0	4402	4538	136
calendar	offline	Stochastic	6919	5818	5818	5482	5482	0
crossed	forward	D2	3683	0	0	227	323	96
crossed	forward	Stochastic	3683	1362	1362	1091	1091	0
crossed	offline	D2	6919	0	1	1376	1391	15
crossed	offline	Stochastic	6919	2491	2491	2030	2030	0

Table 42. Retrospective annual performance on the two-estimator common finite-bound cohort, stratified using the complete reference. Match means that the observed lower bound equals the complete reported H30 within 10^{-8} °C; excess means the reference exceeds that bound. This outcome-defined stratum is unavailable at deployment. A station-year can occur in both strata under different masks. Intervals in brackets resample connected groups with 2,000 repetitions and are descriptive under the sampled archive, not independent-climate guarantees.

Family	Setting	Stratum	Method	n	Groups	MAE	Coverage	Group interval
calendar	forward	Match	D2	3125	17	0.000	100.0%	[100.0, 100.0]
calendar	forward	Match	Stochastic	3125	17	0.000	99.9%	[99.6, 100.0]
calendar	forward	Excess	D2	549	15	0.191	88.3%	[84.6, 93.2]
calendar	forward	Excess	Stochastic	549	15	0.151	87.8%	[84.1, 93.5]
calendar	offline	Match	D2	5891	18	0.000	99.9%	[99.6, 100.0]
calendar	offline	Match	Stochastic	5891	18	0.000	99.9%	[99.8, 100.0]
calendar	offline	Excess	D2	1028	16	0.122	90.6%	[86.8, 92.9]
calendar	offline	Excess	Stochastic	1028	16	0.120	83.1%	[79.8, 88.7]
crossed	forward	Match	D2	1537	17	0.012	99.3%	[98.9, 99.8]
crossed	forward	Match	Stochastic	1537	17	0.025	94.4%	[89.1, 99.3]
crossed	forward	Excess	D2	2146	17	0.378	83.2%	[72.6, 93.0]
crossed	forward	Excess	Stochastic	2146	17	0.345	81.6%	[72.2, 91.7]
crossed	offline	Match	D2	2787	18	0.008	99.3%	[98.9, 99.6]
crossed	offline	Match	Stochastic	2787	18	0.005	99.7%	[99.5, 99.9]
crossed	offline	Excess	D2	4132	18	0.285	90.7%	[88.5, 93.6]
crossed	offline	Excess	Stochastic	4132	18	0.289	87.9%	[84.6, 91.3]

Table 43. Exploratory source-only annual interval expansion on two-method common cases. Supported cases have at least three disjoint source groups and a positive empirical correction; unavailable and zero-correction regions remain in the denominator. Before and after use exactly the same supported cases. The source-group-weighted empirical correction has no finite-sample or climate-transfer guarantee; it was explored after inspecting the principal results. Coverage is percent, width and interval score are in °C.

Family	Setting	Method	Eligible	Supported	Coverage	Width	Interval score
calendar	forward	D2	3676	0	–	–	–
calendar	forward	Stochastic	3676	0	–	–	–
calendar	offline	D2	6926	0	–	–	–
calendar	offline	Stochastic	6926	0	–	–	–
crossed	forward	D2	3683	92	100.0→100.0	0.532→0.543	0.532→0.543
crossed	forward	Stochastic	3683	92	100.0→100.0	0.555→0.567	0.555→0.567
crossed	offline	D2	6919	0	–	–	–
crossed	offline	Stochastic	6919	0	–	–	–

Table 44. USGS count-only preflight for the additional fixed metadata roster. The 2025 extension was decided after the 2010–2024 results. A missing count is shown as –; eligibility is unknown when a missing 2025 count could change the four-year decision. No temperature values are shown.

Station	2010–2024 complete years	2025 approved days	Extended count decision
02469100	0	208	Does not pass
12514400	0	–	Does not pass
15200280	1	262	Does not pass
15236900	0	–	Does not pass
15292400	0	–	Does not pass
15302812	3	–	Unknown
15493000	3	214	Does not pass
15515500	0	207	Does not pass
15580095	0	134	Does not pass
15908000	0	95	Does not pass

9 Source-provider dependence of whole-archive models

The 42 whole-archive target stations belong to providers with recorded reuse paths, but source-model training has additional provenance. Recomputing the source role assignments and checking them against all 13 saved seed-zero model bundles shows that 13 estimator roles and 6 interval-calibration roles have positive-weight LUBW source cases. Source cases are repeated masks; their counts are not independent observations. Positive source weights establish model dependence but do not resolve the legal status of any particular derivative. Individual source, model, target, mask and reviewer-access rights still require confirmation. A post-hoc candidate-provider strict-region refit is reported separately and does not replace the main results.

Table 45. LUBW cases with positive source-role weights, not independent station counts. Groups with bootstrap multiplicity zero do not contribute.

Mode	Region	Estimator case rows	Interval calibration case rows
forward	foen	383	0
forward	gkd_bayern	287	0
forward	usgs_huc03	383	0
forward	usgs_huc04	96	0
forward	usgs_huc10	191	96
forward	usgs_huc17	191	0
offline	foen	288	311
offline	gkd_bayern	862	2732
offline	syke	1030	311
offline	usgs_huc03	3475	0
offline	usgs_huc04	1461	168
offline	usgs_huc10	1150	0
offline	usgs_huc17	1246	2732

10 Candidate-provider, strict-region source sensitivity

After observing the original results and LUBW source dependence, we refitted all 13 whole-archive source models on seven candidate providers with recorded reuse paths. Role assignments and bootstrap draws were recomputed using the filtered data, and every group occurring in the target region was excluded from source roles. An independent role recount found no positive-weight target-region source case or post-2009 forward source case. Original target masks and daily features were reused. All 3,320 archive settings, masks and the unchanged plug-in baselines were checked. Removing providers, changing source assignments and strict regional exclusion jointly change the analysis; the result is not a causal attribution to one of them. The filtered inputs remain processed study data, not provider-native raw records. This after-the-fact sensitivity neither establishes legal redistribution rights nor independent climate transfer, and it does not replace the original analysis. A descriptive connected-group bootstrap resamples the same groups for both source versions and both estimators. It does not establish independent climates or a population transfer guarantee. The corresponding paired MSE differences and bootstrap intervals are supplied with the candidate-source summary.

Table 46. Trend RMSE in °C per decade on the four-way common cases: both methods and both source-model versions must be available. Repetitions are not independent climates.

Family	Mode	Cases	Groups	D2 orig.	Class. orig.	D2 cand.	Class. cand.
calendar	forward	701	17	0.06178	0.05134	0.06033	0.05822
calendar	offline	835	18	0.00873	0.01124	0.00890	0.01119
crossed	forward	738	17	0.16381	0.14658	0.15806	0.15600
crossed	offline	838	18	0.02780	0.03208	0.02868	0.03195

Table 47. Nominal 95% D2 trend-interval results on cases where both versions are available. Width is in °C per decade; this comparison is descriptive and the selected cases may differ from the four-way point cohort.

Family	Mode	Cases	Cover orig.	Cover cand.	Width orig.	Width cand.
calendar	forward	820	96.0%	95.7%	0.116	0.115
calendar	offline	840	95.0%	95.2%	0.018	0.018
crossed	forward	820	88.0%	87.6%	0.450	0.408
crossed	offline	840	94.0%	92.9%	0.089	0.086

Table 48. Symmetric MSE decomposition on cases with all four retention-by-estimator cells available under both versions. Fractions divide the mean retention contribution by mean total improvement; they are not causal or universal.

Family	Mode	Estimator	Cases	Groups	Original fraction	Candidate fraction
calendar	forward	D2	820	17	0.799	0.783
calendar	forward	classical	701	17	0.744	0.767
calendar	offline	D2	840	18	0.887	0.891
calendar	offline	classical	835	18	0.931	0.930
crossed	forward	D2	807	17	0.939	0.937
crossed	forward	classical	725	17	0.925	0.931
crossed	offline	D2	840	18	0.930	0.933
crossed	offline	classical	838	18	0.942	0.942

11 Annual uncertainty under candidate-source refits

The follow-up compares the same target cases where both methods and both source versions have finite seven-quantile distributions and a finite observed bound. The original 24 annual summary cells were independently reproduced on their original cohorts. Target identities, bounds and connected groups agree between versions. The decomposition by whether the complete reference exceeds the visible bound uses the withheld reference and is unavailable at decision time. Candidate-versus-original differences are post hoc; changed source providers, role assignments and strict region exclusion cannot be separated. The quoted CRPS uses the stated normal-spacing tail continuation, and no new calibration or independent climate validation is established. For forward controlled gaps with hidden peaks, the paired candidate-minus-original classical coverage difference is -3.4 percentage points with descriptive connected-group bootstrap interval [-8.6, 2.0]. Removing FOEN or Bavaria from the fixed evaluation cases changes that difference to +0.3 or -8.9 percentage points, respectively. Regions were not refitted in this deletion, and connected groups are not proven independent climates.

Table 49. Annual 90% interval comparison on the same four-way common cases. Hidden peak means that the complete reported H30 exceeds the observed bound; this is an outcome-defined stratum. Widths are in °C. Repeated masks are not independent climates.

Family / mode	Stratum	Method	Cases	Groups	Cover old	Cover new	Width old	Width new
calendar / forward	all	D2	3674	17	98.3%	97.9%	0.118	0.135
calendar / forward	all	classical	3674	17	98.1%	97.9%	0.088	0.093
calendar / forward	hidden	D2	549	15	88.3%	86.3%	0.690	0.680
calendar / forward	hidden	classical	549	15	87.8%	86.7%	0.566	0.610
calendar / offline	all	D2	6919	18	98.5%	98.5%	0.095	0.093
calendar / offline	all	classical	6919	18	97.4%	97.7%	0.066	0.069
calendar / offline	hidden	D2	1028	16	90.6%	90.8%	0.573	0.578
calendar / offline	hidden	classical	1028	16	83.1%	85.1%	0.427	0.434
crossed / forward	all	D2	3683	17	90.0%	89.2%	0.826	0.734
crossed / forward	all	classical	3683	17	87.0%	87.3%	0.693	0.650
crossed / forward	hidden	D2	2146	17	83.2%	81.9%	1.299	1.161
crossed / forward	hidden	classical	2146	17	81.6%	78.2%	1.121	1.078
crossed / offline	all	D2	6919	18	94.1%	92.8%	0.730	0.701
crossed / offline	all	classical	6919	18	92.7%	92.4%	0.660	0.639
crossed / offline	hidden	D2	4132	18	90.7%	88.6%	1.171	1.127
crossed / offline	hidden	classical	4132	18	87.9%	87.4%	1.068	1.037

12 Single-year source expansion for annual intervals

After reviewing prior annual and trend results, we used single-year masked source cases from the candidate models' rule-calibration groups. These groups do not enter the estimators, original interval calibration or risk-learning roles; every target-region group is excluded. Some groups contributed to the separate risk-rule component; the pool is not an unopened independent dataset. Forward source cases are from before 2010. For each family, mode, region and estimator we estimate an equal-group-weighted empirical 90th percentile of nonnegative interval misses. A correction is applied only with at least three available source groups and a positive radius. This rule is exploratory, and the source cases hide a single year rather than an entire degraded archive. It neither establishes group exchangeability nor identifies a calibrated joint trend distribution. At most eight forward groups are assigned to this role, below the nine needed for a finite one-maximum-per-group 90% rank; the group-maximum diagnostic was not applied. Re-prediction of the 52 source support cells, verification of individual target intervals and a separate two-method support-intersection check passed. Group-resampling and leave-region-out summaries are supplied; they do not prove independent climates. Unsupported cells and unchanged all-case baselines remain visible. The source group-mass empirical percentile is not a finite-sample coverage guarantee.

Table 50. Exploratory interval expansion: same target cases must be supported for both methods. Baseline and expanded scores use the identical supported subset; all-case baselines are retained separately in the supplied CSV. Hidden-peak strata depend on the complete reference. Widths are in °C.

Family / mode	Stratum	Method	Eligible	Supported	Groups	Cover before	Cover after	Width before	Width after
calendar / forward	all	D2	3674	0	0	–	–	–	–
calendar / forward	all	classical	3674	0	0	–	–	–	–
calendar / forward	hidden	D2	549	0	0	–	–	–	–
calendar / forward	hidden	classical	549	0	0	–	–	–	–
calendar / offline	all	D2	6919	0	0	–	–	–	–
calendar / offline	all	classical	6919	0	0	–	–	–	–
calendar / offline	hidden	D2	1028	0	0	–	–	–	–
calendar / offline	hidden	classical	1028	0	0	–	–	–	–
crossed / forward	all	D2	3683	3411	14	88.7%	89.9%	0.705	0.755
crossed / forward	all	classical	3683	3411	14	86.6%	89.1%	0.608	0.696
crossed / forward	hidden	D2	2146	1982	14	81.0%	82.9%	1.110	1.171
crossed / forward	hidden	classical	2146	1982	14	76.9%	81.3%	1.009	1.115
crossed / offline	all	D2	6919	0	0	–	–	–	–
crossed / offline	all	classical	6919	0	0	–	–	–	–
crossed / offline	hidden	D2	4132	0	0	–	–	–	–
crossed / offline	hidden	classical	4132	0	0	–	–	–	–

13 Trend sensitivity to annual interval expansion

After examining the source-only annual correction and the earlier trend results, we propagated it through the original 256-draw normal-spacing quantile representation. The endpoint-only scheme moves the 5th and 95th quantiles while holding the five interior quantiles fixed. The tapered scheme also shifts the 10th, 25th, 75th and 90th quantiles by 0.8, 0.4, 0.4 and 0.8 times the radius, respectively; the median and complete-year values are unchanged. Both schemes use the same adjusted 90% endpoints and respect observed lower bounds. Their interior and normal-spacing outer tails differ. The same independent, auxiliary shared-parameter and residual-block rank matrices are reused within each scheme; no calibrated joint distribution is implied. Among 820 forward controlled-gap station–repetition settings, 760 belong to regions with source support for both methods. Seventy-three of these lack at least one annual distribution, and two have no incomplete year, leaving 685 common nontrivial settings. Original saved trend intervals are reproduced before adjustment. The target selection and calibration remain retrospective. Connected-group paired bootstrap and fixed-prediction leave-one-region-out summaries are supplied. They are descriptive, use only 14 target groups and do not establish independent climates. Annual 90% endpoint correction does not identify the inner or outer distribution, cure annual undercoverage, guarantee trend coverage, or resolve source rights. The underlying case records and rank arrays remain local pending permission.

Table 51. Nominal 95% finite-record trend interval comparison on the same 685 station–repetition cases, 38 stations and 14 connected groups. Width and interval score are in °C per decade; repeated masks are not independent climates.

Method	Rank construction	Marginal scheme	Coverage	Width	Interval score
D2	independent	baseline	87.9%	0.382	0.742
D2	independent	endpoints	89.3%	0.409	0.736
D2	independent	tapered	88.9%	0.403	0.735
D2	shared parameter	baseline	88.0%	0.379	0.741
D2	shared parameter	endpoints	89.2%	0.405	0.730
D2	shared parameter	tapered	89.2%	0.399	0.733
D2	residual blocks	baseline	88.2%	0.385	0.721
D2	residual blocks	endpoints	89.8%	0.411	0.708
D2	residual blocks	tapered	89.5%	0.406	0.707
classical	independent	baseline	83.4%	0.344	0.912
classical	independent	endpoints	86.9%	0.387	0.856
classical	independent	tapered	85.5%	0.376	0.866
classical	shared parameter	baseline	82.2%	0.342	0.922
classical	shared parameter	endpoints	86.3%	0.383	0.863
classical	shared parameter	tapered	85.0%	0.373	0.872
classical	residual blocks	baseline	82.6%	0.349	0.901
classical	residual blocks	endpoints	86.4%	0.391	0.845
classical	residual blocks	tapered	85.7%	0.381	0.853

14 Late-start USGS count-only screen

After earlier candidate screens, we fixed an exploratory metadata-only roster of 13 sites with series beginning in 2017–2021 and ending during or after 2025; 11 of the 24 initial candidates had known previous contact and were excluded. The 2017–2025 approved annual sample-count screen confirmed at least four complete calendar years for 0 sites. This count does not prove that other sites lack such years: unreturned annual statistics are unknown. The roster was not prospective and was availability-selected after earlier screens. Counts may include estimated values and do not establish complete subdaily coverage, river suitability, network independence, measurement quality, or permission to reuse data. No daily temperature values or reconstruction errors were used.

Table 52. Late-start USGS count-only candidates. A full year requires an approved annual interval from January 1 to December 31 and 365 or 366 approved daily observations. Missing annual statistics are not interpreted as zero. Network status and all other reference criteria remain unresolved for some or all sites.

Station	Confirmed full years	Returned year statistics	Full years
01376307	0	7	–
01376520	0	7	–
01473470	2	6	2022, 2024
01484100	0	7	–
02327100	2	7	2023, 2024
04010510	3	5	2021, 2022, 2023
04213500	2	6	2022, 2025
11458000	0	1	–
15041200	0	7	–
15219000	3	6	2021, 2023, 2024
15304010	0	0	–
15747000	1	9	2018
255138081321701	2	9	2019, 2023

15 Additional Nechako subdaily reference qualification

The 2019–2024 author archive supersedes its 2019–2021 version. Before opening temperature sequences, published site dates were used to select the only river or creek candidate whose metadata could span all four UTC years 2020–2023; named lakes and a spillway were excluded. The frozen criterion requires exactly 96 unique quarter-hour timestamps per UTC day, finite values, and author flags P or A. The download contains 32 sites; only the selected candidate sequence was opened. No year meets this criterion, so no recovery model was scored. Each year also contains at least one nonfinite value independently of whether other flagged readings are admitted. This site is on the same Nechako River as an earlier BC exploratory case and is not an independent climatic confirmation. The source is Kokoszka et al. (2025), DOI 10.5281/zenodo.15053907, CC BY 4.0.

16 Poudre Water Quality Network reference qualification

A metadata-only screen selected four locations whose source periods span at least four UTC years in 2020–2024. The selected sites share one watershed; older USGS station metadata from the same river had already been encountered, and exact prior-network connections remain unresolved. Before opening temperatures, we fixed a criterion of 96 unique quarter-hour UTC observations per day, finite post-processing values in degrees Celsius and author PASS status. The author metadata describes the unit as degrees C, while the source file uses C; that notation mapping was recorded after inspecting unit labels and before calculating completeness. None of the 20 site-years qualifies; each lacks at least 28 distinct quarter-hour timestamps even before the value and quality filters. Thus no recovery model was scored. Source files contain other sites and parameters that were not qualified here. The authors note that technician calibration records were not collected for 2019–2021; PASS flags do not prove error-free sensors or independent climates. Source: Struthers et al. (2026), DOI 10.4211/hs.521e317e189e498581d0aaa08c3f32bc, CC BY 4.0.

17 Nova Scotia verified workbooks and open-portal comparison

Four river locations were fixed from the provider's historical workbook list before opening their temperature sequences. The workbooks are labelled verified, but their inspected headers do not identify per-hour temperature quality or

Table 53. Frozen P/A flag and 96-quarter-hour qualification of the selected Nechako site. Incomplete days include missing or nonfinite measurements, rejected author flags, duplicates or off-grid timestamps. Author flags are recommended classifications rather than proof of error-free measurements.

Year	Days	Complete	Incomplete	Nonfinite records	Rejected flags
2020	366	341	25	1	1918
2021	365	211	154	6523	13982
2022	365	125	240	21021	21889
2023	365	127	238	17076	20129

Table 54. Author PASS, finite-value and 96-quarter-hour qualification for the four metadata-selected Poudre sites. Missing timestamps are counted independently of quality flags; incomplete days can also contain nonfinite or non-PASS records. These are four sites within one watershed.

Site	Year	Calendar days	Complete days	Missing timestamps	Nonfinite	Not PASS
archery	2020	366	290	28	4441	3200
archery	2021	365	205	11291	2835	2135
archery	2022	365	168	13603	2289	24
archery	2023	365	168	12080	1381	1053
archery	2024	366	207	11428	830	920
cottonwood	2020	366	11	4497	11666	10601
cottonwood	2021	365	0	35040	0	0
cottonwood	2022	365	183	15902	282	22
cottonwood	2023	365	195	11139	481	466
cottonwood	2024	366	216	11424	344	438
riverbend	2020	366	178	28	8852	7992
riverbend	2021	365	227	11283	1469	1111
riverbend	2022	365	190	13604	1335	1174
riverbend	2023	365	188	11996	1193	1230
riverbend	2024	366	210	11433	148	236
salyer	2020	366	129	28	9622	8005
salyer	2021	365	218	11273	1507	1144
salyer	2022	365	88	20270	4531	804
salyer	2023	365	200	11138	1496	1529
salyer	2024	366	217	11346	719	770

estimated-value flags; one header states UTC–04:00 and the remaining headers do not establish a time basis. We count complete reported clock-label years, requiring 24 unique hourly labels with finite numeric cells per calendar day, without claiming these labels correspond to 24 independent physical hours or nonestimated values. The openly licensed portal was accessed only after the workbook completeness screen; it is another release of the same monitoring programme, not an independent sensor. Of 14 station-years with complete labels in both releases, 12 have every numeric value equal within the originally specified 10^{-6} °C tolerance. These include at least four such years at 2 stations. Across 262,667 shared unique numeric labels, 12,810 differ at that tolerance; 1,086 exceed 0.5 °C in a post-hoc diagnostic, with a maximum 4.493 °C. These comparisons do not identify which version is correct, and clock-label matching does not resolve physical time. The original reference screen did not score a reconstruction method; a later, explicitly exploratory amendment is reported separately. The portal source is Government of Nova Scotia, dataset bkfi-mjgw, Open Government Licence – Nova Scotia; legacy workbook reuse remains unresolved.

18 Nova Scotia reported-clock exploratory masking

The preceding qualification did not establish strict reference quality. After inspecting availability and two-release agreement, a separate unregistered exploratory amendment fixed two sites and eight station-years before any model prediction errors were computed. Each selected year has 24 finite reported hourly labels per day and agrees numerically across the two releases of the same monitoring programme. Time zone, estimated-value status and point-level quality remain unresolved. The reported daily value is the mean of the 24 reported hourly values; it is not guaranteed to represent a physical civil day or an error-free temperature. The fixed global source model and source-only interval calibration were reused. Each year was masked separately with 30-, 60- and 90-day controlled gaps, using the other three selected years for local history; no empirical-calendar family or whole-archive degradation was evaluated. NASA POWER forcing uses local solar time and coordinates rounded to 0.01 degree. The adjacent-normal tail continuation is an assumption for CRPS. The 144 cases are repeated masks at two sites and eight station-years, not 144 independent watersheds. An independent arithmetic audit

recomputed reference H30, bounds, masks and reported scores. In two post-hoc hidden-value probes, adding 5 °C to hidden days changed the true H30 but did not change decision features or predictions when identically serialized controls were compared. Those two probes are not an exhaustive leakage proof.

19 Nova Scotia reported-day boundary sensitivity

After inspecting the exploratory results, we tested hypothetical daily bins shifted by −1, 0 and +1 hour relative to the recorded clock labels. We required all original eight station-years and the adjacent boundary hours to be unique, finite and equal within 10^{-6} °C across the two releases; missing or disputed hours were not filled. Daily water inputs and local leave-one-year-out models were rebuilt for each shift, with the original masks, global source model, source calibration and NASA POWER local-solar-time forcing held fixed. All 144 masks are available across all three shifts for both methods; the zero-shift replay agrees with the original outputs. Across the eight reported reference years the largest H30 change is 0.0023 °C, while a post-hoc casewise diagnostic finds a largest D2 point-prediction change of 0.094 °C. This does not identify the true timezone or explain the cause of prediction changes; hourly values, quality and meteorological day alignment remain uncertain. The two sites and repeated masks do not justify a population-level ranking or independent-climate inference.

20 Source role audit for partitioned forward models

We re-derived the seed-zero bootstrap weights and checked them against the roles saved in 98 model bundles for the four evaluated source splits. The selected held test stations and their recorded connected groups did not enter any source role. However, in the before-2010 split, the role function defines held groups from test-period cases; training-only groups in the same named region may therefore remain in the source. The table reports positive-weight same-region source cases, not held-target observations. This is not a uniformly held-region design; the distinction affects the interpretation of forward transfer. It does not establish leakage of target-station hidden values or physical independence across groups.

Table 55. Retrospective 2009–2017 reported-label completeness and release concordance. Equal means that every hourly numeric value in the common complete-label year matches to 10^{-6} °C at the same reported clock label. Quality, estimated-value status, time basis and climate independence are not established.

Station	Workbook	Portal	Both	Fully equal
NS01DL0009	4	5	4	4
NS01ED9111	2	2	1	1
NS01EF0002	4	4	3	3
NS01FB0001	6	6	6	4

Table 56. Reported-clock exploratory controlled-gap scores on identical available cases. Coverage is percent, and MAE, width and interval score are in °C. These are two sites, not independent climatic replications; source quality and time basis remain unresolved.

Method	Cases	MAE	90% coverage	90% width	Interval score
D2	144	0.311	88.2	1.000	1.572
Stochastic	144	0.313	93.1	1.281	1.706

Table 57. Post-hoc descriptive paired results by gap length and site. Counts are repeated masks; group differences are not inferential tests.

Axis	Level	Cases	D2 MAE	Stochastic MAE	D2 coverage	Stochastic coverage
Gap days	30	48	0.247	0.231	89.6	93.8
Gap days	60	48	0.297	0.319	93.8	95.8
Gap days	90	48	0.387	0.388	81.2	89.6
Station	NS01DL0009	72	0.400	0.393	81.9	93.1
Station	NS01FB0001	72	0.221	0.232	94.4	93.1

Table 58. Paired exploratory reported-day boundary sensitivity with identical original masks. Coverage is percent and error, width and interval score are in °C. Local stages are refit; source model, source calibration and forcing are held fixed. Two sites and eight station-years are not independent climatic replications.

Shift (h)	Method	Cases	MAE	90% coverage	Width	Interval score
-1	D2	144	0.310	88.9	1.000	1.555
-1	Stochastic	144	0.313	92.4	1.283	1.704
0	D2	144	0.311	88.2	1.000	1.572
0	Stochastic	144	0.313	93.1	1.281	1.706
1	D2	144	0.315	86.8	0.998	1.572
1	Stochastic	144	0.313	93.1	1.278	1.731

Table 59. Seed-zero before-2010 model roles with positive-weight source cases in the named target region. Bootstrap-weighted counts include repeated role draws and do not represent independent samples.

Region	Role	Groups	Stations	Cases	Weighted cases
nve_hydapi	interval_calibration	2	3	431	695
usgs_huc17	estimator	1	1	140	280

21 Post-hoc strict-region sensitivity

We refitted the two affected before-2010 seed-zero models after removing every source connected group that appears in the named target region. The original assignment and bootstrap multiplicities of other groups were retained; no replacement groups were drawn. Source interval and release thresholds were recalculated from the filtered roles. In the whole-archive experiment, only the three affected HUC17 stations were reanalysed, on the same 120 original masks; all other original results were retained. Each strict run checked the original mask and the absence of hidden target values. Both estimators were compared on the same station–repetition cases available in both versions. For forward controlled gaps this common cohort comprises 738 cases at 41 stations. Original and strict D2 trend RMSE are 0.1638 and 0.1645; the corresponding stochastic RMSE values are 0.1466 and 0.1466 °C per decade.

The future-year release analysis replaces predictions in both affected regions while preserving all target cases and the other regional models. Across 27,846 forward controlled-gap cases, the three-output rule publishes 63.1% and 64.5% in the original and strict comparisons, with 10.9% and 11.3% observed failures among the published values. In HUC17, publication changes from 62.7% to 71.6%, and the selected error rate from 6.0% to 8.2%. The selected sets differ, so the conditional rates cannot be interpreted without retention. The strict refits use fewer source observations; these paired comparisons do not identify the causal effect of an individual group and do not establish external validity or safety. They do not recompute every analysis in this paper under strict regional exclusion. Case counts contain repeated masks and are not independent climates.

The source-role comparison additionally checked held test station identities across all source regions, the positive bootstrap weights and the effective split of classifier and probability-calibration groups. Only the two before-2010 models changed positive source weights; zero-weight role membership still matters for downstream group-list use, so this does not prove every downstream result unchanged.

The release-then-trend pipeline was replayed for all 120 HUC17 settings with the original masks, source-specific strict limits and newly fitted model. Original annual states reproduced for 1,440 station–year records. Across the affected records,

39 observed-bound outputs become numerical estimates; the other observed-bound and estimate states remain unchanged. The original and strict pooled controlled-gap comparisons comprise 820 common station–repetition cases and give released-values-plus-bounds working-likelihood trend RMSE 0.2024 and 0.1986 °C per decade, respectively, with mean working interval widths 3.26 and 3.29. These working intervals include regression variation and do not establish calibrated missingness-only coverage.

In the outcome-defined stratum with an unobserved peak, the 2,146 common forward controlled-gap annual cases have D2 nominal 90% coverage 83.2% originally and 82.9% under the stricter source. This retrospective stratum is unavailable to a decision maker and repeated masks are not independent climate realizations. The affected HUC17 source rule-calibration pool contains at most one eligible connected group before model prediction, below the minimum of three for both annual expansion and task calibration; strict refitting cannot change that source-group upper bound, although the corresponding radius and errors were not recalculated.

For forward controlled gaps, the same 818 station–repetition cases from 17 connected groups have every annual, tail and rank-coupling output in both source versions. Under constant tails, nominal 95% D2 trend coverage is 83.0% originally and 83.3% strictly; normal-spacing tails yield 88.8% and 88.9%, with widths 0.451 and 0.454 and interval scores 0.860 and 0.862. In the separate normal-tailed rank experiment, independent ranks yield 88.0% and 87.9% coverage and residual blocks yield 89.0% and 89.0%. Original tail results reproduced on all 480 affected tail rows. These comparisons use the same random ranks within each tail or coupling experiment, but the tail and coupling experiments use different draws; they do not identify additive causes of undercoverage. Descriptive connected-group bootstrap intervals for strict-minus-original metrics accompany the aggregate results.

For the forward controlled-gap D2 versus stochastic comparison, 738 cases at 17 connected groups are available under both source versions and both estimators. The pooled D2-minus-stochastic MSE difference is +0.005350 originally and +0.005559 strictly; the descriptive connected-group bootstrap interval for the strict result is [+0.000957, +0.010431] $(°C/\text{decade})^2$. The same resampled groups are used for both source versions. Removing Bavaria from these fixed predictions leaves 331

cases in 9 groups and changes the difference from -0.000143 originally to +0.000323 strictly. For empirical calendars the corresponding original and strict differences are -0.000213 and -0.000159 on 332 cases. These small changes can alter the sign of the regional comparison; the leave-one-region-out exercise does not refit source models and does not establish independent climates or significance.

The small controlled-gap stress comparison reanalyses 20 settings for one HUC17 station while retaining other stations' original results. Among 30 common baseline/stress station–repetition cases, original and strict baseline D2 RMSE are 0.0794 and 0.0803; for synchronized donor gaps they are 0.1164 and 0.1159, and for heat-associated gaps 0.0885 and 0.0902 °C per decade. Original scenario metrics and masks reproduce. The heat-associated scenario changes both the identity and potentially the number of masked years, so these are exploratory paired scenario comparisons, not isolated causal effects or independent climate tests.

A separate recalculation of the transductive, same-number numerical-output comparator under the strict source model publishes 17,961 controlled-gap values: three-output and score-only failure rates are 11.26% and 7.15%, respectively. The score threshold uses the target feature distribution and the three-output count; counts and selected cases differ between source versions, and lower-bound information is not held equal.

22 Additional external-reference metadata checks

A metadata-only check of the H.J. Andrews HT004 stream-temperature catalog (`https://andrewsforest.oregonstate.edu/data/datacatalog/HT004`) identified the Lookout Creek gaging station as USGS 14161500 (`https://andrewsforest.oregonstate.edu/data/studies/hf04/hja_gage_ws_descriptions_instrumentation.pdf`). That station is already in the source archive and its recorded connected group contains 14 source stations; different HJA probes require separate identity assessment. No HJA temperature records were downloaded or scored in this check. This is prior-contact evidence, not an estimate of HJA data quality or independent accuracy.

The RivTemp–MFFP Quebec catalog describes validated daily means (`https://datastream.org/en-ca/dataset/09ab9bf0-789b-4487-918e-b5c8cf258225`).

The currently described RivTemp protocol permits daily values when more than 90% of within-day observations are retained (`https://rivtemp.ca/en/data-validation/`); a particular published study instead describes removing a day when an hourly observation is missing (`https://doi.org/10.1016/j.hydroa.2022.100133`). The rule applicable to a specific historical release, the underlying time basis, unique interval completeness and redistribution terms require clarification. We acquired three published RivTemp partner files and screened their activity-date labels without reading temperature values or scoring models. Across 248 MFFP, 132 INRS and 4 Vision H2O location identifiers, none has four complete calendar years by this optimistic label criterion in the checked published periods. The MFFP 2010–2020 screen was fixed before examining dates; the wider MFFP period and the two other partners were explored after that screen. A separate Arrow-based recount reproduced the station–year label counts. Labels do not establish finite or nonestimated measurements, a full 24-hour day, physical station independence or accuracy. No new reference was qualified.

New Jersey continuous-monitoring intake. The NJDEP portal provides downloadable quality-reviewed records (`https://njdep.rutgers.edu/continuous/download/`). The station selector contained 287 entries, of which 156 were labeled freshwater; 86 freshwater entries lacked quality-period boundaries and remain unresolved. After discovering inconsistent water-type labels, a post-hoc check found 26 entries whose quality-period envelopes spanned at least four possible calendar years; all were labeled marine, and 24 had nonzero record counts in at least four queried years. Counts can include different variables and do not prove complete water-temperature observations. We selected 4 river/stream candidates after inspecting metadata and counts, then acquired their 2016–2019 temperature files. All 16 returned station-years lacked some reported date labels, with 37–168 missing dates per station-year. The downloaded CSV row counts agree with the provider count endpoint for all 16 station-years. An independent Arrow/Python implementation reproduced the timestamp counts without examining temperature values or model errors. The files use an EST label, but the real time basis, expected sampling cadence, nonestimated status, individual quality flags and catchment independence remain unresolved. These selected cases do not qualify as complete reference years

under their reported-clock convention; this post-hoc screen does not establish the number of qualifying stations in the full portal.

23 Prior Australian reference exposure

An earlier Australian recruitment stage identified 80 observation-supported candidate station-years from 52 stations. The historical intake record states that its designated F2 formal test was closed; this audit does not run that model or calculate new F2 errors. Subsequently, a separate, earlier-method Australian evaluation used calendar-year cases from the same acquired source archive. All 80 of the candidate station-years appear in that case file; 79 had at least one case in the old evaluation's eligible case set, including all 10 of the 10 originally designated F2 test-role station-years. The prior evaluation contained 30,728 eligible masked cases across its wider roster. These are previously evaluated records for a different pipeline, not unopened target station-years for a new independent test.

A later Australian development analysis used July–June rather than calendar-year windows. Counting physical time overlap rather than matching the year labels alone, 74 of the original 80 calendar station-years intersect a saved development window, of which 73 intersect at least one eligible development case. A separate Arrow/Python recount reproduced these counts. Matching station and interval does not prove that masked days are identical, that the F2 model was evaluated, or that data leaked into source-model training. The original native-sampling criterion allowed 20 observed hours per day and does not establish a complete 24-hour physical reference. These previously opened Australian records cannot supply a fresh independent confirmation of the current pipeline.

24 Independent observed-record arithmetic re-audits

Using the same local canonical float32 daily archive and an independent convolution implementation, we recomputed 2,139 complete reference station-years and 545 incomplete station-years at 96 eligible stations. All 35 witness identities, saved 30-day values and complete-year reference maxima agreed within the stated numerical tolerance. This is an arithmetic re-audit of the same processed archive, not an independent check of provider-native records or sensor error.

Comparing 971,673 finite values from 34 selected project input panels with the canonical archive yielded no missingness or float32-cast discrepancies. The largest absolute rounding difference is 0.00000153 degrees C; 0 witness identities and 0 classifications at the 0.5-degree excess threshold changed. The input panels are processed project files, not provider-native instrument records.

Across 2,700 eligible station-years, independently recalculated annual coverage, available 30-day maxima, July–August completeness and longest missing runs reproduced 210 witness–rule decisions. For each of 35 witness years, increasing its observed values by 5 degrees C did not change the leave-target-year-out warm-season climatology. The old fixed-season fraction remains sourced from saved metadata; these probes do not cover every year or other pipelines.

For the ten additional USGS sites, independent parsing of the saved daily API responses reproduced 8 approved-complete reference station-years and the 9 certified observed windows. Approved estimated values remain in complete-year references, but the certified windows contain no estimated days. The quality-screened choice of window was not separately reconstructed. These records do not establish subdaily completeness, physical measurement accuracy or independent recovery performance.

25 Exploratory British Columbia daily-reference comparison

All 23 author-selected published stations were retained for qualification. Four stations each have four complete calendar years in the published daily series. The original study accepted daily means based on at least 20 hourly values after researcher quality checks; hourly counts, original flags and the day convention are not available in this derivative. The source-water cleaning used water–air comparisons. A complete published year is not equivalent to 24 independently quality-checked hours per day. The 23-station author cohort was selected for data availability and these results are unregistered. The fixed global source model and independent source-role recalibration are used; local reconstruction models are refit after holding out the scored year. Each year is masked separately, with 30-, 60- and 90-day controlled gaps. NASA POWER air forcing uses coordinates rounded to 0.01 degree from the source article; there are no same-panel donor stations. The score treats the published daily mean as reference, not as error-free physical truth. Repeated

gaps in a station-year and the four stations are dependent; three stations lie in the Fraser River basin (Nechako, Thompson and Fraser). These are descriptive results, without population uncertainty or a natural-outage validation. Across 372 masked

Table 60. Controlled-gap annual H30 mean absolute errors in degrees Celsius. Each row has four published-daily reference years; mask counts are not independent observations.

Station	Masks	Plug-in	D2	Stochastic
08FF003	72	0.833	0.475	0.520
08JA017	72	0.241	0.243	0.203
08LF051	72	0.383	0.207	0.220
08MF040	72	0.605	0.298	0.265

cases, independently recomputed 30-day reference maxima, observed-window bounds, missing counts, and 52 point and interval summary metrics agree with stored results within numerical tolerance. As a limited noninterference probe, one 90-day case per station was regenerated after adding 5 degrees C to hidden target values. All 4 probes retained the same decision-time model features and numerical predictions, while the evaluation reference changed. These probes do not establish noninterference for every case, model path, or experiment. The attribution, candidate roster, year-level qualification, individual station summaries and code are included in the local evidence package. Prior-identifier and geographic proximity checks are best effort and cannot prove independence from every previously opened record.

26 Earlier diagnostic figures

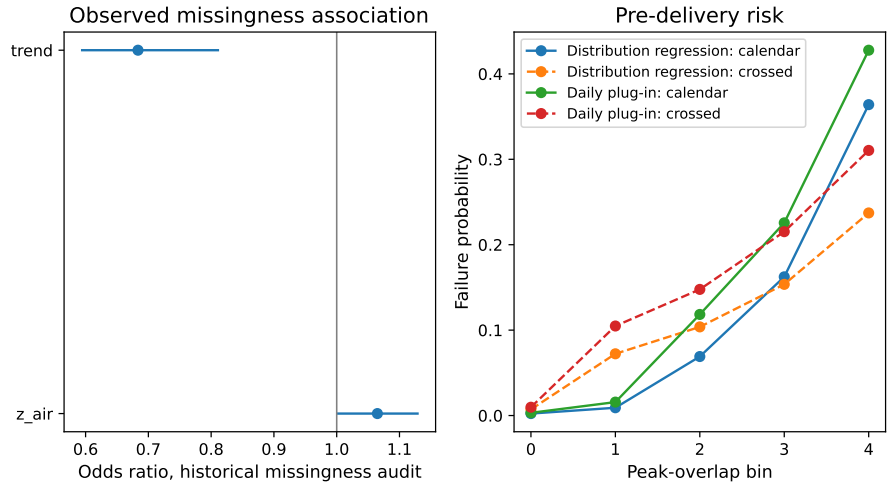


Figure 3. Decision-time risk concentration across peak-overlap bins. Source tables preserve populations, layers and repetition variability.

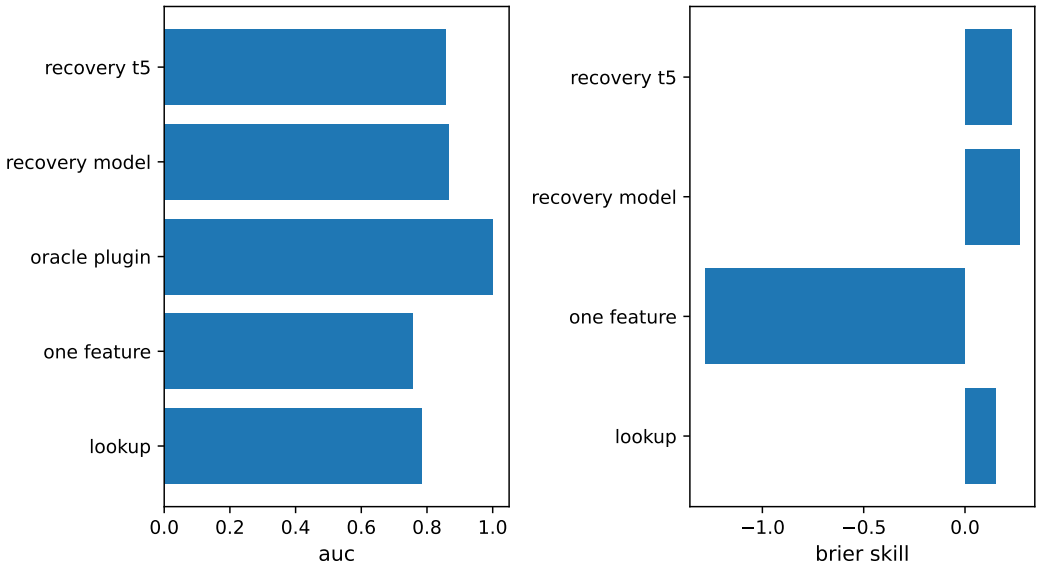


Figure 4. Recovery-model discrimination and Brier skill against simple scores. Ranking does not imply calibration.

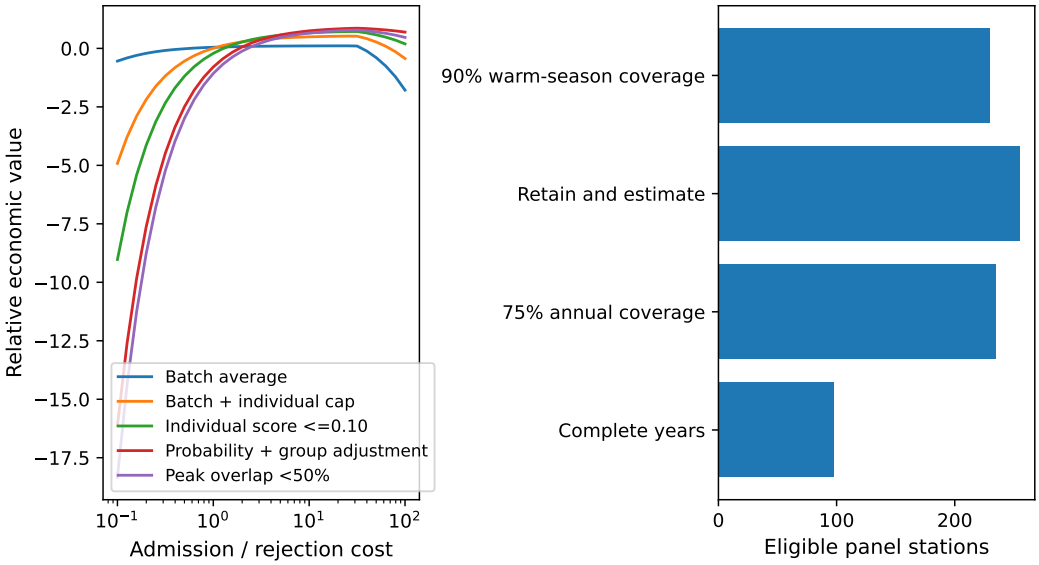


Figure 5. Decision cost sensitivity and station eligibility in the assembled archive.

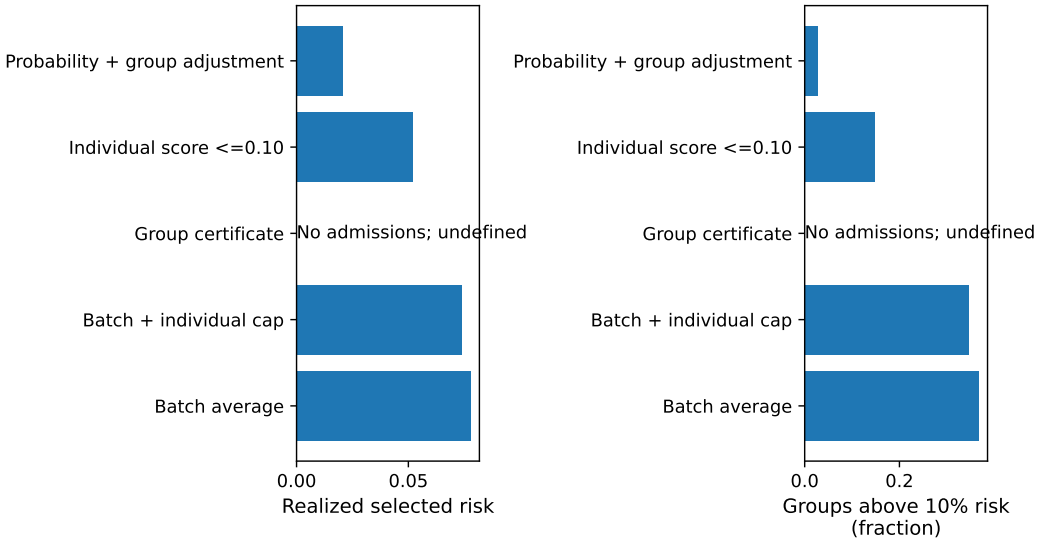


Figure 6. Average risk and the fraction of evaluable groups with risk above 10%, reported separately.

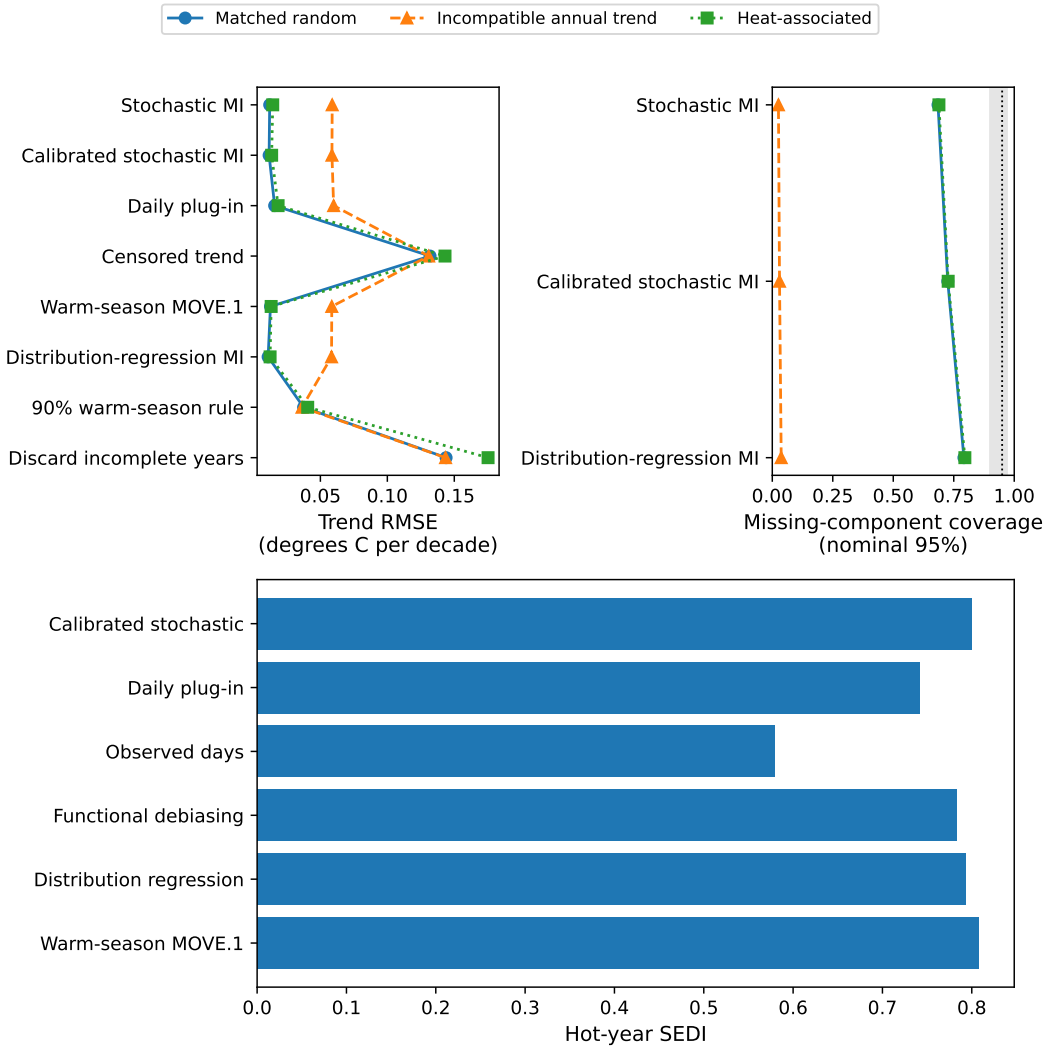


Figure 7. Trend RMSE, missing-component interval coverage and hot-year event skill. The shaded interval is the 90–97% coverage gate for nominal 95% missing-component intervals. Reduced RMSE does not establish calibrated functional uncertainty. The incompatible annual diagnostic differs from the separate full daily refit.